ResearchGate

See discussions, stats, and author profiles for this publication at: https://www.researchgate.net/publication/249547542

Palaeocene–early Eocene inversion of the Phuquoc–Kampot Som Basin: SE Asian deformation associated with the suturing of Luconia

Article *in* Journal of the Geological Society · March 2010

DOI: 10.1144/0016-76492009-039

CITATIONS
39

READS
534

8 authors, including:



Michael Fyhn
Geological Survey of Denmark and Greenland
37 PUBLICATIONS **422** CITATIONS
SEE PROFILE



Stig A. Schack Pedersen
Geological Survey of Denmark and Greenland
91 PUBLICATIONS **1,187** CITATIONS
SEE PROFILE



Lars Ole Boldreel
University of Copenhagen
78 PUBLICATIONS **960** CITATIONS
SEE PROFILE



Lars Henrik Nielsen
Geological Survey of Denmark and Greenland, GEUS
142 PUBLICATIONS **2,132** CITATIONS
SEE PROFILE

Some of the authors of this publication are also working on these related projects:

PhD publication View project

The Triassic-Jurassic boundary: Impact of a Large Igneous Province on the geobiosphere View project

All content following this page was uploaded by Michael Fyhn on 20 December 2017.

The user has requested enhancement of the downloaded file.

Journal of the Geological Society

Palaeocene early Eocene inversion of the Phuquoc Kampot Som Basin: SE Asian deformation associated with the suturing of Luconia

Michael B.W. Fyhn, Stig A.S. Pedersen, Lars O. Boldreel, Lars H. Nielsen, Paul F. Green, Phan T. Dien, Luong T. Huyen and Dirk Frei

Journal of the Geological Society 2010; v. 167; p. 281-295
doi:10.1144/0016-76492009-039

Email alerting service click here to receive free email alerts when new articles cite this article

Permission request click here to seek permission to re-use all or part of this article

Subscribe click here to subscribe to Journal of the Geological Society or the Lyell Collection

Notes

Downloaded by Koebenhavns Universitetsbibliothek Nord on 1 March 2010

© 2010 Geological Society of London

Lyell Collection

Journal of the Geological Society, London, Vol. **167**, 2010, pp. 281–295. doi: 10.1144/0016-76492009-039.

Palaeocene–early Eocene inversion of the Phuquoc–Kampot Som Basin: SE Asian deformation associated with the suturing of Luconia

MICHAEL B. W. FYHN[1,2]*, STIG A. S. PEDERSEN[2], LARS O. BOLDREEL[1], LARS H. NIELSEN[2], PAUL F. GREEN[3], PHAN T. DIEN[4,5], LUONG T. HUYEN[4] & DIRK FREI[2]

[1]*Department of Geography and Geology, University of Copenhagen, Øster Voldgade 10, DK-1350, Copenhagen K, Denmark*

[2]*Geological Survey of Denmark and Greenland, GEUS, Øster Voldgade 10, DK-1350, Copenhagen K, Denmark*

[3]*Geotrack, Melbourne, 37 Melville Road, Brunswick West, Victoria 3055, Australia*

[4]*Hanoi University of Mining and Geology, Dong Ngac, Tu Liem, Hanoi, Vietnam*

[5]*Vietnam Petroleum Institute, Dong Da, Truong Chinh, Hanoi, Vietnam*

**Corresponding author (e-mail: mbwf@geus.dk)*

Abstract: The little explored Cambodian and Vietnamese Phuquoc–Kampot Som Basin is a Late Jurassic to Early Cretaceous foreland basin developed in response to the build-up of a palaeo-Pacific magmatic arc. A combination of seismic data, well data and outcrop geology complemented by fission track and U/Pb analysis is used to unravel the basin history. This reveals a hitherto unknown earliest Palaeogene basin inversion associated with the Luconian suturing to SE Asia and the shutdown of palaeo-Pacific subduction underneath SE Asia. The Phuquoc–Kampot Som Basin and the Khorat Basin in Thailand constitute the erosional remnants of a larger basin that covered large parts of SE Asia in Late Mesozoic time, and subsequently became segregated during earliest Palaeogene inversion and erosion. Inversion was focused along the several hundred kilometres long Kampot and Khmer–Chanthaburi fold belts that confine the Phuquoc–Kampot Som Basin and merge with the Mae Ping and the Three Pagodas fault zones. These connections, together with local NW–SE-trending sinistral transpressional faults offshore, indicate a link between initial SE Asian left-lateral strike-slip faulting and the Luconian suturing. The separation between the once unbroken Khmer–Chanthaburi Fold Belt and the Phetchabun Fold Belt in Thailand suggests a 50–100 km Cenozoic left-lateral offset across the Mae Ping Fault Zone.

The Sundaland core of SE Asia arose from the amalgamation of smaller continental fragments throughout the mid-Phanerozoic (Fig. 1). The main features of the accretion history are fairly well described, although the exact timing of continent collision is debated (Hutchison 1989; Mitchell 1993; Lovatt Smith *et al*. 1996; Metcalfe 1996, 1998; Stokes *et al*. 1996; Lepvrier *et al*. 1997, 2004, 2007; Lacassin *et al*. 1998; Sone & Metcalfe 2008; Barber & Crow 2009). The Late Mesozoic to earliest Cenozoic development is less known, although a series of basins formed during this period, which may record the regional coeval tectonic development (Fig. 2). Indeed, the early Palaeogene accretion of Luconia onto SE Asia has been documented only in Sarawak in western Borneo despite its proximity to Vietnam (Fig. 1; Benard *et al*. 1990; Hutchison 1996; Honza *et al*. 2000).

Eocene–Oligocene rifting of the Cenozoic basins along the Vietnamese margin was associated with major left-lateral shearing across narrow fault zones transecting the region (Tapponnier *et al*. 1986; Rangin *et al*. 1995; Fyhn *et al*. 2009*a*,*b*,*c*). The activity of these fault zones has generally been linked with the southeastward displacement of the region caused by the Indian–Eurasian collision during the mid-Cenozoic (Tapponnier *et al*. 1986; Lacassin *et al*. 1993, 1997). However, an earlier onset of sinistral shearing related to Cretaceous to early Palaeogene Tethys subduction along the western SE Asia margin or the accretion of western Myanmar onto SE Asia has been documented (Morley 2004; Watkinson *et al*. 2008; Searle & Morley in press). S-type magmatism and metamorphism along the northwestern rim of Sundaland combined with moderate basin inversion in the northern Khorat Basin suggests that Cretaceous–early Palaeogene convergence along western SE Asia had a major impact on the regional tectonic development (Charusiri *et al*. 1993; Mitchell 1993; Lovatt Smith *et al*. 1996; Barley *et al*. 2003; Mitchell *et al*. 2007; Searle *et al*. 2007; Barber & Crow 2009).

Knowledge of the Indochinese evolution from after the Late Palaeozoic–Early Mesozoic accretion of Sundaland until the onset of Cenozoic deformation associated with the Himalayan orogeny is fragmentary. Hence, the potential link between Indochinese basin evolution and Late Mesozoic–earliest Cenozoic plate convergence along the eastern margin of Sundaland, similar to that described from the western part of Sundaland, is little explored.

This paper investigates this 'missing link' between Late Mesozoic basin formation and mid-Cenozoic deformation associated with the Himalayan orogeny. Based on a study of the Late Jurassic to Early Cretaceous Phuquoc–Kampong Som Basin, we present a model for the basin development and inversion of the Late Mesozoic Indochinese basins. The model includes a reinterpretation of the late-stage accretion of SE Asia in addition to the onset age and mechanism of left-lateral shearing within southern Indochina. The study is based on analysis of *c*. 18 000 km of multichannel 2D seismic data from offshore south Vietnam tied to wells (Fig. 3). The seismic analysis was combined with outcrop studies on the SW Vietnamese and Cambo-

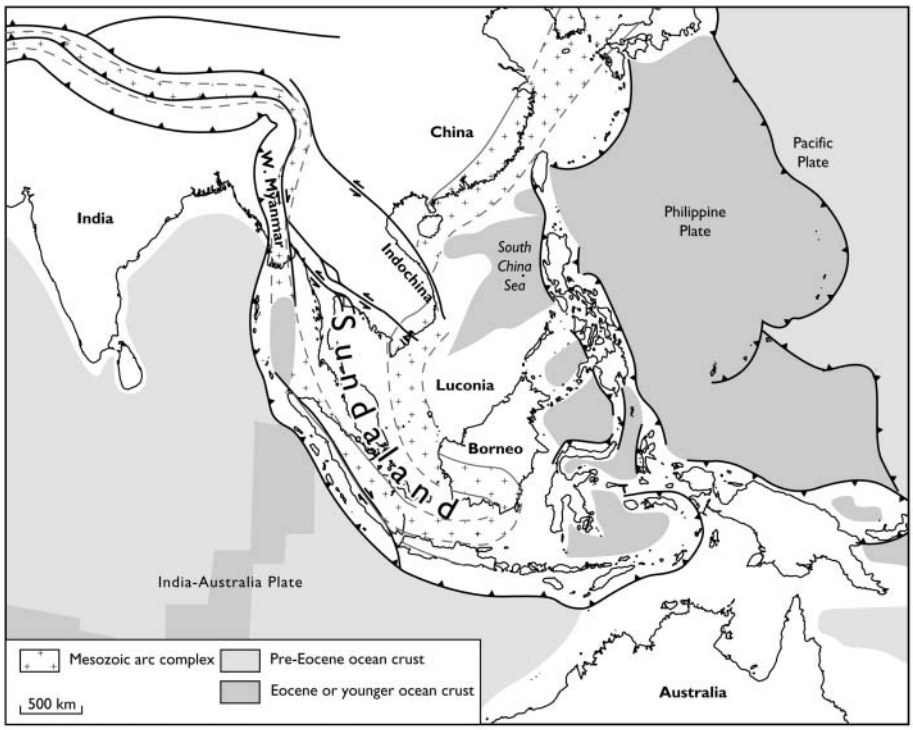


Fig. 1. Structural outline of the SE Asian region illustrating the main Cenozoic structures of the area. A Mesozoic magmatic arc outlined Sundaland, which together with the post-Eocene ages of most ocean basins to the east of it indicates a significantly different Mesozoic regional outline.

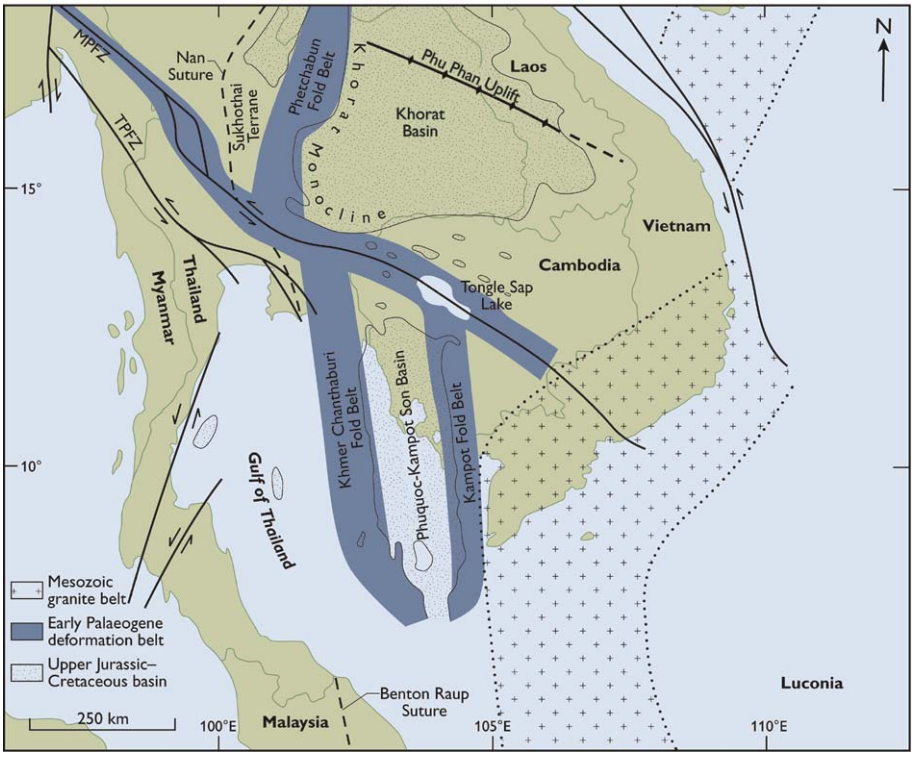


Fig. 2. Index map of Indochina with selected structural elements. Two major fold belts transect the region and connect to the Mae Ping Fault Zone (MPFZ) and the Three Pagodas Fault Zone (TPFZ). The four structural belts outline the boundaries of large parts of the Late Jurassic–Early Cretaceous basins. The Khmer–Chanthaburi Fold Belt has been offset from the Phetchabun Fold Belt across the Mae Ping Fault Zone, which suggests 50–100 km of Cenozoic left-lateral motion across the fault zone.

dian mainland and on nearby islands. A fully cored, nearly 500 m deep well was drilled through Lower Cretaceous (Aptian) sediments on Phuquoc Island, complementing the outcrop study. Exhumation ages and timing of magmatism were assessed using apatite fission track analysis (AFTA) and U/Pb zircon dating respectively.

Summary of the regional geology

Accretion of Sundaland

A series of latest Permian–Triassic Indosinian sutures outline smaller Gondwana-derived continental fragments that constitute the Sundaland core of the SE Asian ‘promontory’ (Figs 1 and 2;

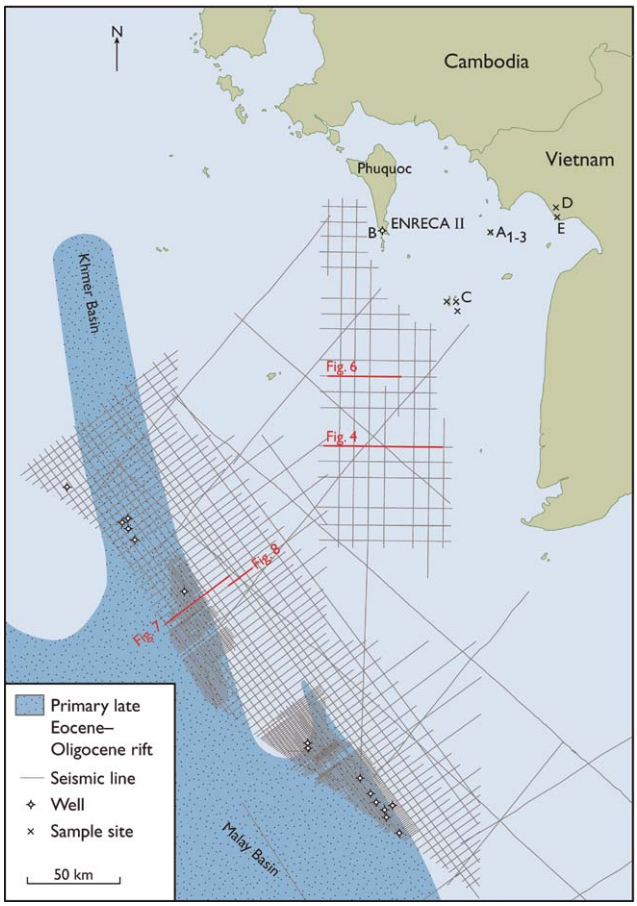


Fig. 3. Seismic grid across the study area with available wells and sample sites. Illustrated seismic lines are marked in bold. AFTA ages and mean track lengths; A1, 58.8 ± 3.8 Ma and 13.93 ± 0.18 μm; A2, 89.6 ± 12.6 Ma and 13.3 ± 0.55 μm; A3, 59.1 ± 7.9 Ma and 13.26 ± 0.32 μm; B, 51.3 ± 3.6 Ma and 13.23 ± 0.19; C, 62.7 ± 5.9 Ma and 12.94 ± 0.49 μm; D, 52.8 ± 3.3 Ma and 13.71 μm; E, 53.4 ± 2.7 Ma and 13.61 ± 0.17 μm.

Hutchison 1989; Metcalfe 1996; Sone & Metcalfe 2008). The Nan and Benton Raub Sutures outline some of the principal Indosinian suture zones of Sundaland. The sutures are considered to enter the Gulf of Thailand west of the study area as roughly north–south-trending lineaments paralleling the trend of the Cenozoic rifts underlying the sea (Fig. 1).

Following the main amalgamation, an Andean-type margin was established along the fringe of Sundaland (e.g. Metcalfe 1996). Convergence of the palaeo-Pacific (Panthalassa) along the east coast of Asia resulted in the creation of a magmatic arc parallel to the continental margin during the Mesozoic. Remnants of the eastern magmatic arc are found in the Schwaner mountains of Borneo and as Jurassic–Cretaceous igneous complexes forming the basement of offshore Tertiary basins between Borneo, the Malayan Peninsula and south Vietnam (Fig. 2; Katili 1973; Haile *et al*. 1977; Williams *et al*. 1988; Hutchison 1989, 1996). The arc can be traced from the offshore basins across south Vietnam to south China and farther to the NE (Jahn *et al*. 1976; Areshev *et al*. 1992; Rangin *et al*. 1995; Zhou & Li 2000; Li *et al*. 2004; Nguyen *et al*. 2004; Thuy *et al*. 2004). Cenozoic extension and translation has dislocated the magmatic belt and differential block rotation contorted its original shape (Haile *et al*. 1977; Williams *et al*. 1988; Leloup *et al*. 1995; Fuller *et al*. 1999). In particular, the deviating trend of the magmatic-arc system observed in the Schwaner Mountain Belt of Borneo (roughly WNW–ESE) relative to that farther north (roughly north–south to NE–SW) probably resulted from a 50–90° counter-clockwise rotation of Borneo since the Cretaceous, indicated by palaeomagnetic investigations (Fig. 1; Haile *et al*. 1977; Schmidtke *et al*. 1990; Fuller *et al*. 1999). From the Late Mesozoic until the middle Eocene, the known west palaeo-Pacific plates drifted to the NW to NNW with respect to Asia (Engebretson *et al*. 1985; Koppers *et al*. 2001; Seton & Müller 2008), which most probably controlled subduction along the Asian margin. Hence, subduction to the NW to NNW along the magmatic arc of Borneo is compatible with a large subsequent counter-clockwise rotation of the island, which produced its present outline. However, because of the restricted preservation of Late Mesozoic west Pacific oceanic lithosphere, including potential intra-oceanic divergent, convergent and translational plate boundaries, reconstruction of the direction of Mesozoic and earliest Cenozoic Pacific plate motions is somewhat speculative (Engebretson *et al*. 1985; Koppers *et al*. 2001; Honza & Fujioka 2004; Smith 2007; Seton & Müller 2008).

The Jurassic to earliest Palaeocene ages recorded in I-type arc-related igneous rocks of SE Indochina indicate that convergence operated along this part of the margin during this period (Table 1; Areshev *et al*. 1992; Rangin *et al*. 1995; Hoa 1996; Tri 1999; Tinh 1998; Trang 1998; Thang 1999; Nguyen *et al*. 2004; Thuy *et al*. 2004). Subduction beneath the southeastern margin terminated during the Palaeocene to late Eocene and is recorded by the Sarawak orogeny in western Borneo (Hutchison 1996; Honza *et al*. 2000). The orogeny coupled with the halt of subduction in this part of the region are viewed as a consequence of the collision of Sundaland and the Luconian Block (Fig. 2; Hutchison 1996).

More recently Hall (2009) and Hall *et al*. (2009) suggested cessation of palaeo-Pacific subduction already during the Late Cretaceous as a result of suturing of the Luconia–Dangerous Grounds Microcontinent to SE Asia. This interpretation was based mainly on: (1) the assumption of cessation of arc magmatism around middle Late Cretaceous time, and (2) the suggestion of Moss (1998) that the Cretaceous to Eocene Rajang–Embaluh Group in northern Borneo was not part of an accretionary prism associated with subduction underneath Borneo, as otherwise suggested (Haile 1968, 1974, 1994; Hutchison 1973, 1989, 1991, 1996, 2005; Katili 1973; Hamilton 1979; Holloway 1981; Williams *et al*. 1988, 1989; Benard *et al*. 1990; Tan & Lamy 1990; Tongkul 1991; Hazebroek & Tan 1993). In the central northernmost part of Kalimantan, Moss (1998) found no evidence for overall northwards-younging, scraped-off tracts within the Rajang–Embaluh Group, or of strong deformation or metamorphism of the Rajang–Embaluh Group. Consequently, the Rajang–Embaluh Group was suggested to consist of sediments deposited in a remnant ocean basin (Moss 1998). However, all of the above-mentioned features have been documented in Sarawak and other parts of Borneo by, for example, Hutchison (1996), Omang & Barber (1996) and Honza *et al*. (2000). Although volcanism seems to have peaked during Cenomanian–Turonian time, arc volcanism did not terminate prior to the end of the Cretaceous in Borneo (Hutchison 1996) and in the early Palaeocene in Vietnam (Table 1; Hoa 1996; Tinh 1998; Thang 1999). An accretionary setting for the Rajang–Embaluh Group probably existed until that time (Hutchison 1996; Honza *et al*. 2000). Consequently, we infer that subduction continued along the Borneo–Vietnam margin until around early Palaeocene time.

Table 1. *Compilation of Middle Jurassic to earliest Palaeogene radiometric ages of south Vietnamese intrusive rock*s

Radiometric age (Ma)	Igneous complex	Reference
183 ± 3	VC	1
182	VC	1
178 ± 5	?	2
177 ± 2	VC	1
166 ± 4	VC	1
159 ± 5	?	2
158 H	DC	1
157	DQ	1
157 ± 3	VC (?)	3
155 ± 4	VC	1
155 ± 3 J	VC	1
153 ± 4	DQ	1
153	DQ	1
149 ± 5	?	2
146	VC	1
144	DQ	1
143 ± 2 B	HK	1
141 ± 1	VC	1
140	DQ	1
135 ± 4	?	2
134	DQ	1
131 ± 3 F	DQ	1
128 ± 3	DC	1
126 ± 3	DC	1
121 J	VC	1
121	DQ	1
121 ± 5 E	DQ	1
119 ± 4 I	VC	1
119 ± 2	DC	1
117 ± 3	DC	1
117 ± 2	DQ	1
116 ± 5	DQ	1
115 I	VC	1
113 ± 8	DQ	1
112 ± 2	DQ	4
112	DQ	1
111 ± 3	DC	1
111 ± 11	DQ	1
110 ± 1	CN	1
110	TN	1
109 ± 5	?	2
108 ± 3	?	2
108 ± 4	?	2
106 J	VC	1
106 E	DQ	1
105 ± 5	?	2
104 E	DQ	1
104 ± 2 H	DC	1
100 ± 3 B	HK	1
100 ± 2	DQ	4
100 ± 2	CN	1
100 ± 2	DC	1
99 ± 2	DQ	1
99 ± 4 A	CN	1
98 ± 3	DC	1
98 ± 1	DC	5
98	DC	1
98	CN	1
97 C	DC	1
97 ± 1 I	VC	1
97 ± 2 D	DC	1
97 ± 3	?	2
97 ± 3	DC	1
97 ± 9	DQ	1
96 ± 1	CN	4
96 ± 1	CN	4
96 ± 10	DQ	1
96 ± 2	DC	1
96 ± 2	?	3
95±1	DQ	1
95 ± 1	DC	5
95 ± 1	DC	5
95 ± 1	CN	1
94 ± 1	CN	4
94 ± 2	DQ/CN (?)	3
94 ± 4	CN	1
92 ± 1	DC	4
91 ± 1	DC	4
89 ± 1	DC	4
87 ± 2	DC	6
86 ± 3	CN	1
84 ± 2 C	CN	1
84 ± 3 A	CN	1
83 ± 3	DC	1
82 ± 3 F	DQ	1
82 ± 8	DC	1
79 ± 2	DC	1
78 ± 1	DC	1
78 ± 4 D	DC	1
77 ± 3	DC	1
71 ± 1 A	CN	1
71	DQ	1
70	DQ	1
70	DQ	1
69 ± 3 G	DC	1
62 ± 2	PR	1
60 ± 1	DC	1

(*continued*)

Table 1. (*continued*)

VC, Van Can Complex; DC, Deo Ca Complex; DQ, Dinh Quan Complex; HK, Hon Khoai Complex; CN, Ca Na Complex; TN, Tay Ninh Complex; PR, Phan Rang Complex. 1, Map series 1996–1999 (Hoa 1996; Trang 1998; Thang 1999) (K–Ar ages measured on monomineralic biotite, hornblende and feldspar); 2, Areshev *et al*. (1992) (K–Ar ages measured on monomineralic biotite); 3, Rangin *et al*. (1995) (K–Ar ages measured on whole-rock samples); 4, Nguyen *et al*. (2004) (U–Pb ages on zircon and titanite, Pb–Pb ages on zircon, and Rb–Sr ages on biotite, K-feldspar and plagioclase); 5, this study (U–Pb zircon ages); 6, Lasserre *et al*. (1970) (K–Ar biotite ages). Letter pairs and triples A–J denotes samples collected at or near the same locality.

On the opposite eastern side of Sundaland, smaller continental or arc fragments including the western half of Myanmar accreted onto SE Asia during Cretaceous to earliest Palaeogene time (Mitchell 1993; Metcalfe 1996; Barley *et al*. 2003; Mitchell *et al*. 2007; Searle *et al*. 2007; Barber & Crow 2009; Hall *et al*. 2009). Morley (2004) and Watkinson *et al*. (2008) identified Cretaceous to earliest Palaeogene transpression across western Sundaland forced by either Tethys subduction or accretion of western Myanmar onto SE Asia. However, transpression across the region has more commonly been regarded as a result of the subsequent India–Eurasia collision (Tapponnier *et al*. 1986; Leloup *et al*. 1995, 2001; Lacassin *et al*. 1993, 1997).

Late Jurassic–Cretaceous Sundaland basins

A number of Late Jurassic to Cretaceous and earliest Cenozoic basins are situated across large parts of Sundaland (Koopmans 1968; Gobett & Hutchison 1973; Rishworth 1974; Khoo 1977; Harbury *et al*. 1990; Mouret *et al*. 1993; Heggemann *et al*. 1994; Lovatt Smith *et al*. 1996; Racey *et al*. 1996; Lovatt Smith & Stokes 1997; My *et al*. 2002; Dien *et al*. 2008). Although the sediments of these basins are grouped into various locally defined groups and formations, their remarkably uniform stratigraphy has led to speculations of a common origin (e.g. Koopmans 1968; Rishworth 1974; Khoo 1977; Heggemann 1994). Indeed, a single vast Late Mesozoic basin, subsequently split up by erosion and

lateral shearing, has been suggested (Tapponnier *et al*. 1986; Mouret *et al*. 1993; Heggemann *et al*. 1994).

To the north, the Khorat Basin occupies larger parts of eastern Thailand and the bordering areas of Laos and Cambodia (Fig. 2). This basin has been suggested to be a molasse basin associated with the Indosinian orogeny (Hutchison 1989), a thermal sag basin following Triassic rifting, a foreland basin, most probably associated with uplift along the Anamitic Fold Belt and along Sundaland suture zones (Lovatt Smith *et al*. 1996), or a combination of foreland flexuring and thermal collapse (Cooper *et al*. 1989). The Late Jurassic to Cretaceous formation of the basin and the regional uniform thickness indicates that one of the two latter models is the most likely (Heggemann *et al*. 1994; Lovatt Smith *et al*. 1996; Carter & Moss 1999; Carter & Bristow 2003). The basin comprises the Khorat Group of Late Jurassic to Albian age and the Late Cretaceous to earliest Palaeogene Maha Sarakham Formation (Lovatt Smith *et al*. 1996; Racey *et al*. 1996; Stokes *et al*. 1996; Lovatt Smith & Stokes 1997). During the Late Cretaceous, the Khorat Basin experienced mild inversion and reorganization of the depositional pattern suggested to be related to the coeval accretion of west Myanmar onto western Sundaland (Lovatt Smith & Stokes 1997).

Apatite fission track ages around 40–60 Ma (middle Eocene–Palaeocene) in eastern Thailand and neighbouring Laos document a widespread exhumation event during the earliest Cenozoic (Mouret *et al*. 1993; Lovatt Smith *et al*. 1996; Upton 1999). Increasingly older parts of the Khorat Group are exposed toward the west along the Khorat Monocline, which flanks the Phetchabun Fold Belt to the east. In the central part of the fold belt the Upper Mesozoic section has been completely removed as a result of deep erosion. The monocline continues along the southern margin of the Khorat Basin straddling the Thai–Cambodian border. South of the monocline in Cambodia the erosion level is comparable with that in the Phetchabun Fold Belt (Gustavson Associates 1991). Although Neogene deposits within the Tongle Sap Lake Basin conceal most of the pre-Tertiary geology, scattered outcrops of the Khorat Group as well as older rocks north of the Tongle Sap Lake demonstrate pronounced post-Early Cretaceous erosion (Tien 1991; Vysotsky *et al*. 1994).

South of the Tongle Sap Lake, a similar erosional pattern exists. Here an Upper Jurassic–Lower Cretaceous unit termed the Phu Quoc Formation in Vietnam and Bokor or Cam Pong Formation in Cambodia crops out as erosional remnants together with older sediments and intrusions surrounded by Quaternary sediments (Gustavson Associates 1991; Tien 1991; Vysotsky *et al*. 1994).

Only limited available information exists on the geology of Cambodia and SW Vietnam, and the Phuquoc–Kampot Som Basin has been interpreted as a rift basin or as a foreland basin (Vysotsky *et al*. 1994; Dien *et al*. 2008). Geological maps show the basin flanked to the east by a north–south-trending belt of patchy Palaeozoic to Triassic sediments, metasediments and intrusive rocks that again fringe the Jurassic to earliest Palaeogene magmatic arc farther to the east (e.g. Tien 1991). The Upper Jurassic to Lower Cretaceous succession is most completely exposed along the Elephant Mountains in Cambodia and on the Vietnamese Phuquoc Island paralleling the western margin of the belt. To the west, the Phuquoc–Kampot Som Basin is bordered by a monocline along the Thailand–Cambodia border, comparable with the western margin of the Khorat Basin.

In addition to the distribution of Jurassic–Cretaceous deposits onshore Thailand and Indochina, patches of similar deposits, buried underneath younger Cenozoic sediments in the Gulf of Thailand, have been reported from Thailand (Fig. 1) (Pradidtan & Dook 1992; Morley *et al*. 2004). These patches may represent erosional remnants comparable with those observed onshore.

Farther to the south on the Malayan Peninsula, variably preserved Upper Jurassic to Lower Cretaceous strata form comparable sedimentary units to the Khorat and the Phuquoc groups (Koopmans 1968; Gobbett & Hutchison 1973; Rishworth 1974; Khoo 1977; Hutchison 1989). As with the Khorat and the Phuquoc groups, the Malayan equivalents were deposited in a mainly non-marine setting and are dominated by alluvial or fluvial sediments with subordinate volcanic and volcanoclastic rocks (Hutchison 1989; Harbury *et al*. 1990). The Malayan sedimentary accumulations have tentatively been suggested to be rift fills (Gobbett & Hutchison 1973; Harbury *et al*. 1990). NNW–SSE-trending folding and reverse faulting in addition to moderate tilting in places (Gobbett & Hutchison 1973; Hutchison 1989; Harbury *et al*. 1990) document a post-depositional tectonic event, as does the deep erosion comparable with that observed in Cambodia and on Phuquoc Island. The Mesozoic deposits continue to the north, offshore from the Malayan Peninsula, and erosional remnants sporadically underlie the late Eocene to Recent rift and sag basins of the Gulf of Thailand in Malaysian waters (Hutchison 1989; Ngah 2000).

The Phuquoc–Kampot Som Basin

The Phuquoc–Kampot Som Basin forms an elongated, more than 500 km long sediment-filled depression extending from south-western Cambodia in the north to the central southern part of the Gulf of Thailand (Fig. 2). Geological maps of the region (Tien 1991; Vimuktanandana 1999) together with seismic data suggest that the basin is as an up to *c*. 150 km wide belt with the basin axis located approximately along latitudes 103–104°.

Based on available descriptions and the authors' investigations of the Phu Quoc Formation from outcrops on Phuquoc Island and the Cambodian equivalent, the Bokor or the Cam Pong Formation, as well as interpretation of *c*. 500 m of continuous cores, we suggest that this up to 3–4 km thick unit is assigned to a common group (Gustavson Associates 1991; Vysotsky *et al*. 1994; My *et al*. 2002; Linh 2003; Dien *et al*. 2008). The outcrops on Phuquoc Island are extensive and well mapped (Linh 2003). In addition, the 500 m of continuous core with wire-line logs is a candidate for the type section for the group and are available for future studies at the Vietnam Petroleum Institute in Ho Chi Minh City. The cores were taken from the ENRECA II well, drilled at the southern tip of Phuquoc Island, which encountered an Aptian succession.

It is beyond the scope of this paper to define the proposed new lithostratigraphic group formally following the recommendations of Salvador (1994). In this study we use the informal term 'the Phuquoc group' to encompass the Phu Quoc and the Bokor–Cam Pong formations.

The studied Barremian to Aptian basin fill is dominated by continental deposits consisting of laterally continuous fluvial cross-bedded sandstones interbedded with subordinate lacustrine and flood plain mudstones. Shallow-marine sandstones containing *Diplocraterion*, *Skolithos* and *Thallassinoides* burrows form a minor part of the succession. The fluvial transport direction on Phuquoc Island varied considerably, as indicated by the orientation of foresets. However, the content of rhyolite-dominated volcanic clasts, generally in the range of *c*. 10%, suggests the coeval volcanic arc located to the east as the primary upland area (Dien *et al*. 2008).

The Khorat and the Phuquoc basins have been suggested to be a once continuous large basin, later split by left-lateral strike-slip

movements and/or focused erosion (Tapponnier *et al*. 1986; Mouret *et al*. 1993; Heggemann *et al*. 1994). Although knowledge of the pre-Neogene geology of central Cambodia is restricted at present, a common Late Mesozoic basin history seems probable given the striking similarity in age together with the depositional and erosional style of the sediments in the two basins.

Offshore seismic data reveal the Upper Jurassic to Lower Cretaceous Phuquoc group as an up to *c*. 2 s TWT (two-way travel time) thick seismic mega-sequence corresponding to a thickness of *c*. 3–4 km as indicated by seismic stacking velocities and average acoustic velocities measured in the ENRECA II well. The thickness of the Phuquoc group is governed mainly by the amount of erosion along the top-Mesozoic angular unconformity that caps the group (Fig. 4). In addition, gentle internal wedging causes a stratigraphic thickening toward the magmatic arc (Fig. 4). This wedge-like geometry, combined with the relative position and comparable timing of the basin and the magmatic arc, suggest that the studied part of the Phuquoc–Kampot Som Basin constitutes the preserved foredeep of a retro-arc foreland basin that formed in response to the build-up of the magmatic arc (e.g. DeCelles & Giles 1996; Naylor & Sinclair 2008, and references therein). The obvious lack of syndepositional rifting together with the high content of locally sourced coarse-grained material are in accord with this interpretation.

Basin inversion

Structural style and distribution

Exposed Upper Jurassic to Lower Cretaceous strata cropping out on Phuquoc Island generally show a 10–20° inclination towards

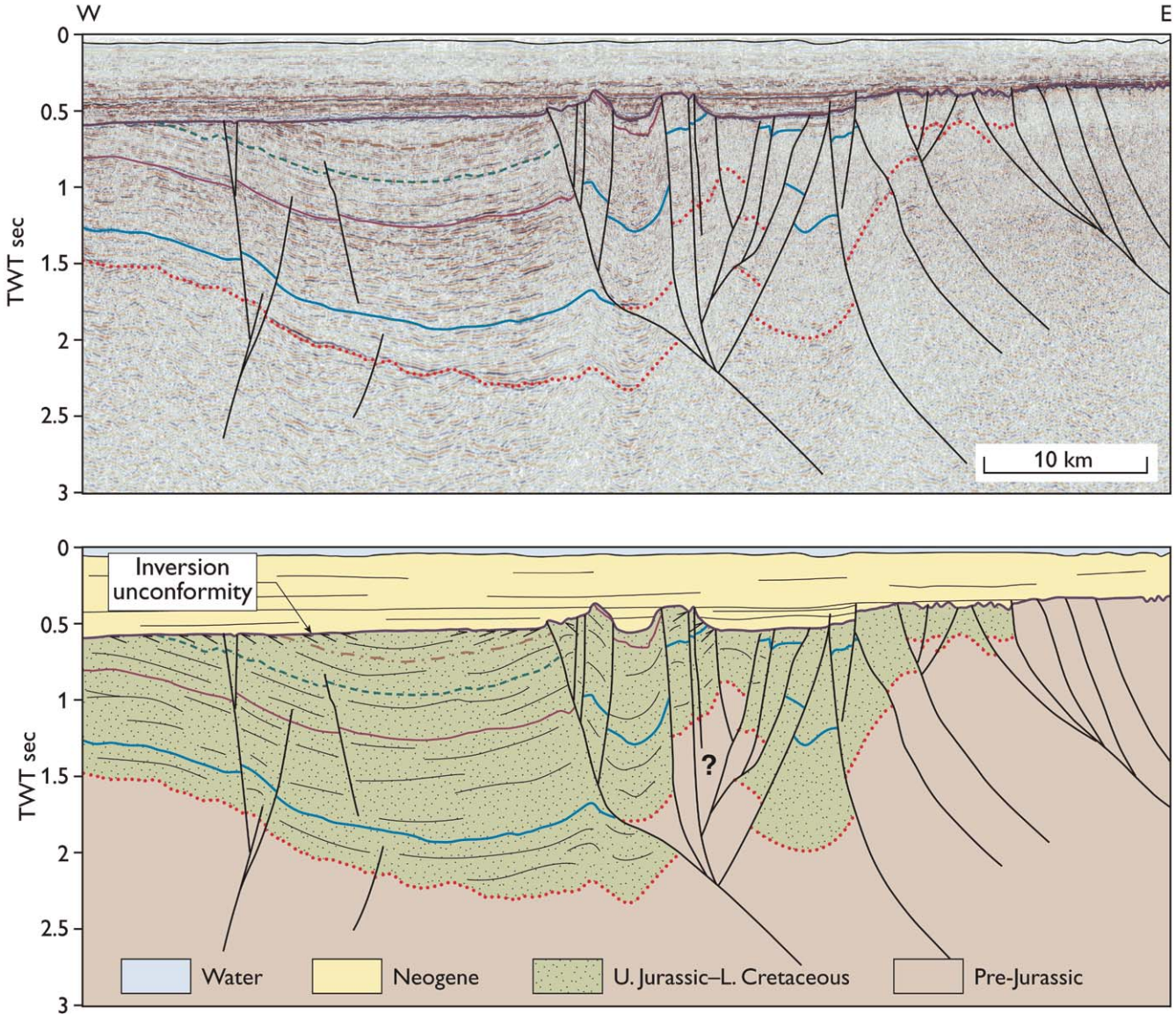


Fig. 4. Seismic transect illustrating intensified structuring toward the Kampot Fold Belt in the east. The Upper Jurassic–Lower Cretaceous Phuquoc group is significantly deformed and truncated at the base-Neogene unconformity, whereas the Neogene sequence is virtually unaffected. A slight internal wedging within the Upper Jurassic–Lower Cretaceous succession indicates stronger subsidence toward the coeval magmatic arc to the east, which suggests a Late Jurassic–Early Cretaceous foreland basin setting.

the west and SW, suggesting a slight post-depositional tectonic tilt (Linh 2003, and authors' own observation). The Cambodian part of the Phuquoc group possesses a comparable tilt, suggesting that the associated tectonic event affected a larger region.

Seismic data reveal distinct contractional faults and folds that are cut off along the top-Mesozoic angular unconformity (Fig. 4). The structural complexity increases towards a north–south-trending deformational belt to the east that cross-cuts the entire study area (Figs 2 and 5). Approximately north–south verging imbricated thrusts and associated folds deform the pre-Cenozoic successions, although the intense deformation within the belt renders detailed seismic mapping difficult (Fig. 4). In general, increasingly older successions subcrop the Mesozoic–Cenozoic boundary in the belt towards the east as a result of eastward intensified structuring. The belt represents the offshore continuation of a more than 100 km broad belt exposed onshore (Tien 1991). The belt will hereafter be referred to as the Kampot Fold Belt after the south Cambodian city. The Phuquoc group crops out in the up to *c*. 1 km high Elephant Mountains and on Phuquoc Island straddling the western part of the belt. Triassic and older sediments or metasediments in addition to intrusive and extrusive rocks crop out in patches along the eastern part of the belt (Tien 1991; Hoa 1996). The Kampot Fold Belt can be traced as far north as a few tens of kilometres south of the Tongle Sap Lake and transects the entire study area offshore covered by seismic data, and thus extends for more than 600 km (Fig. 2). Farther to the east the Kampot Fold Belt borders the Jurassic to earliest Palaeogene magmatic arc, which is occasionally exposed in the southernmost parts of Vietnam and Cambodia.

Fieldwork in the southern onshore part of the Kampot Fold Belt in Vietnam and on islands to the south of the mainland confirms the presence of a generally NNW–SSE- to north–south-trending fold-and-thrust system, although intense weathering and dense vegetation limit the exposures. Deformed Late Jurassic to earliest Palaeogene acidic igneous rocks along with associated tuffs and agglomerates crop out between Triassic and older sedimentary and metasedimentary successions in the eastern part of the Kampot Fold Belt. The igneous rock complexes represent the westernmost part of the SE Vietnamese magmatic arc, with associated agglomerates deposited in limited piggyback basins. Farther to the east, comparable intrusive rocks crop out as

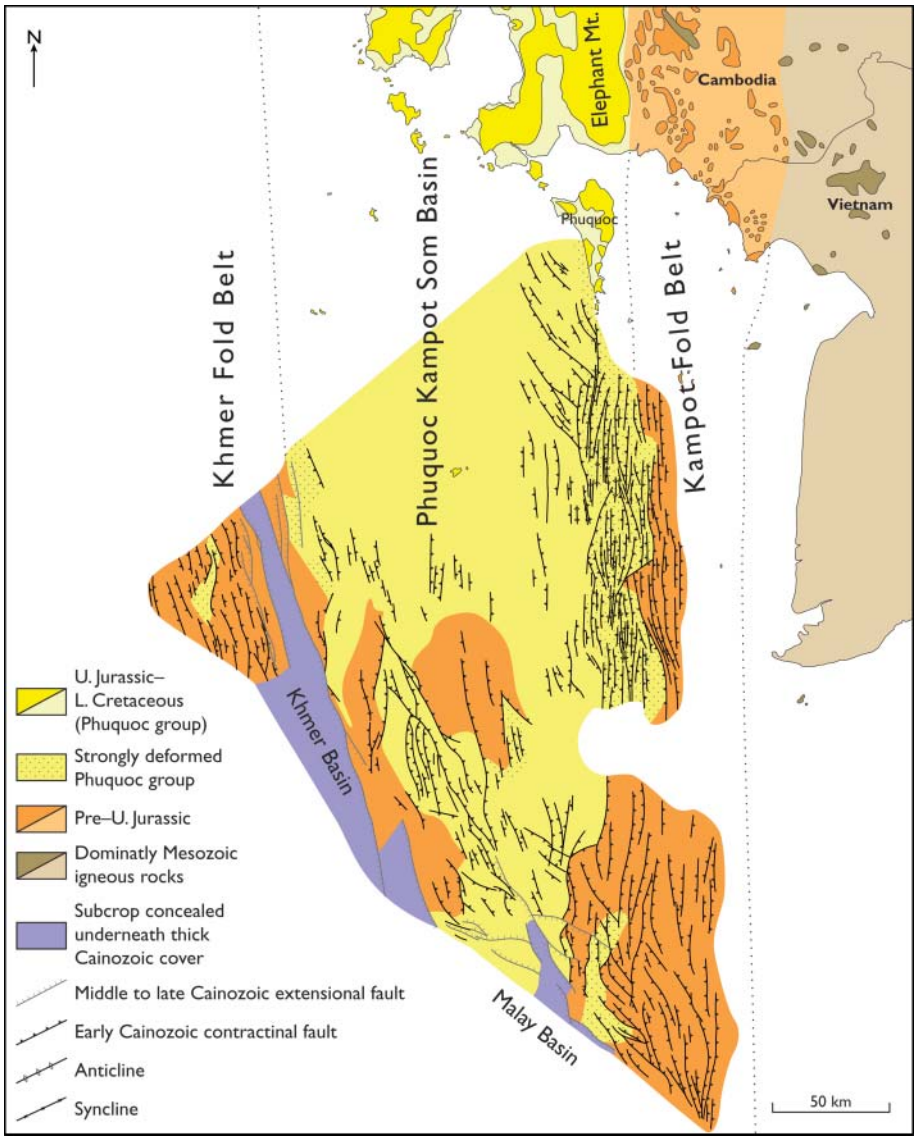


Fig. 5. Map of the subcrop pattern at the top-Mesozoic unconformity that delineates the southern part of the Phuquoc–Kampot Som Basin. Zones of intense thrusting and faulting outline the Kampot and the Khmer fold belts that confine the outline of the Phuquoc–Kampot Som Basin erosionally. Simplified onshore pre-Quaternary outcrops are indicated, outlining the onshore continuation of the Phuquoc–Kampot Som Basin, the Kampot Fold Belt and the SE Indochina Mesozoic magmatic arc. Late Eocene–Oligocene rifting reactivated older contractional crustal fabric and downfaulted part of the top-Mesozoic unconformity below conventional seismic resolution.

isolated hills surrounded by Quaternary alluvium and as islands in the gulf area, constituting the southernmost exposed part of the magmatic arc of Indochina. These intrusive rocks have been deformed by north–south- to NNW–SSE-trending thrust faults, similarly to the Phuquoc group and the eastern part of the deformational belt.

The Kampot Fold Belt represents a strongly eroded orogenic belt that stretches more than 600 km from central Cambodia to the central part of the Gulf of Thailand. South of the available seismic grid the fold belt appear to continue beneath thick Tertiary deposits in the Malay Basin (Fig. 5).

Offshore, the top-Mesozoic unconformity is characterized by a distinct relief across the western flank of the Kampot Fold Belt as a result of a particularly resistant interval in the Phuquoc group repeatedly subcropping the unconformity because of folding and faulting (Fig. 6). A comparable feature has been noted in Thailand, where the Berriasian–Barremian Phra Wihan Formation of the Khorat Group commonly caps the crests of high-lying areas along escarpments as a result of its relative competence and resistance to erosion compared with other stratigraphic intervals (Upton 1999). In the Phuquoc–Kampot Som Basin the offshore relief reappears onshore in the up to >500 m high mountains of Phuquoc Island and in the even higher Cambodian Elephant Mountains. In addition, mid- and late Cenozoic extensional fault movements have contributed to the relief in varying degrees.

The Phuquoc group thins both to the south and the western part of the area as a result of erosion, and in places has been completely removed. Part of the erosion is due to rift-shoulder uplift associated with mid-Cenozoic extension in the Malay and the Khmer basins, documented by extensional faulting that transects the Phuquoc group and terminates at the base of or dies out within the Neogene post-rift succession (Fig. 7). However, dramatic thickness variation of the Phuquoc group occurs across constrictional faults, demonstrating that earlier orogenic uplift was a dominant factor controlling the distribution and thickness of the Jurassic–Cretaceous succession in the southern and western part of the area, as it is also farther to the NE.

Orogenic structuring increases toward the Khmer Basin, along the central part of which the degree of deformation is comparable with that observed across the Kampot Fold Belt farther to the east. A second NNW–SSE-trending deformational belt (hereafter named the Khmer Fold Belt) thus confines the distribution of the Phuquoc group to the west (Figs 1 and 5). The belt strikes along the axis of the Khmer Basin toward the coasts of SE Thailand, where a boundary comparable with the eastern basin margin is outlined on geological maps of Thailand and Cambodia (Vimuktanandana 1985; Tien 1991).

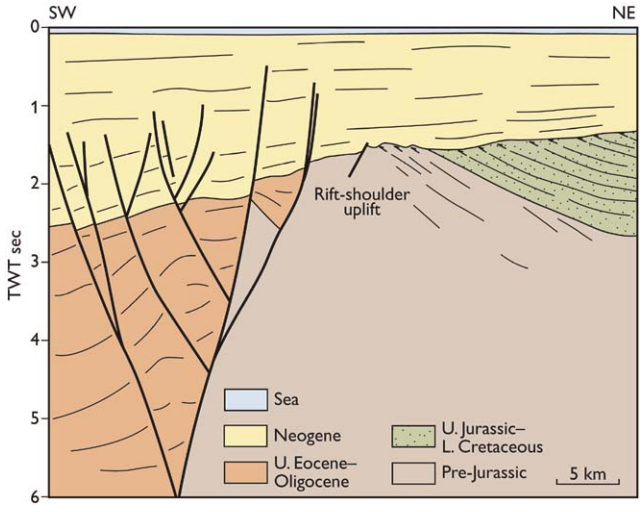


Fig. 7. A pronounced rift-shoulder uplift associated with middle or late Eocene to Oligocene rifting is suggested by the truncation of the pre-Cenozoic succession towards mid-Cenozoic grabens. Erosion related to this rift event influenced a greater part of the region, as suggested by AFTA data.

Fault trends and associated deformational styles

Two contractional fault trends dominate in the study area, trending north–south to NNW–SSE and NW–SE to WNW–ESE, respectively. North–south- to NNW–SSE-trending thrust

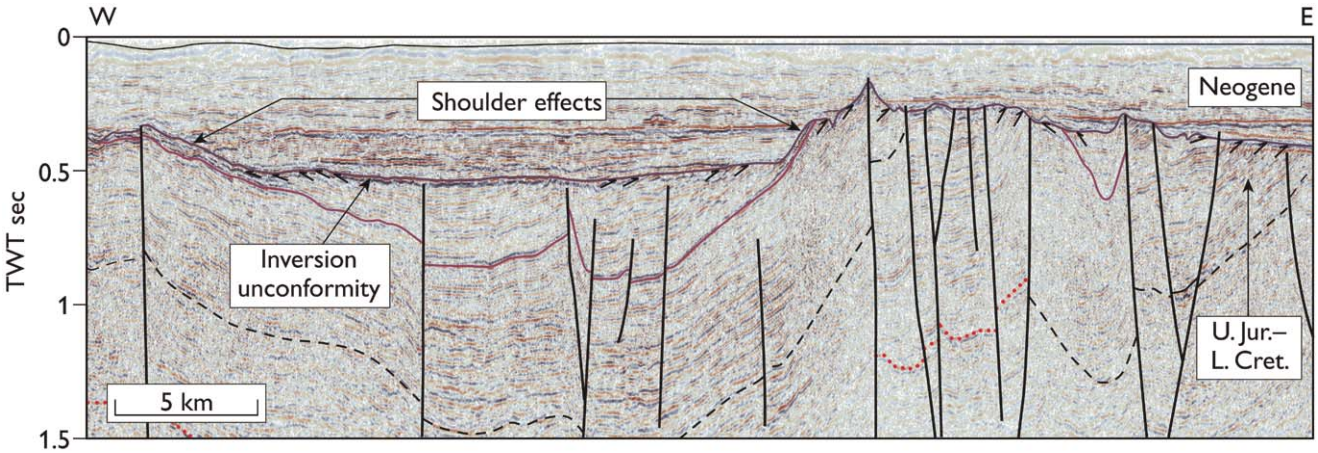


Fig. 6. A distinct unconformity caps the Mesozoic succession and developed in response to basin inversion. The subcrop of a restricted competent stratigraphic interval has resulted in numerous buried ridges, the topography of which reappears onshore. In Thailand outcrops of the Phra Wihan Formation (Berriasian–Berremian) frequently show comparable features (Upton 1999), probably forming the stratigraphic equivalent to the competent interval of the buried shoulders offshore. The topography of the top-Mesozoic unconformity located in the eastern half of the section forms the buried offshore part of the Elephant Mountains–Phuquoc Island mountain chain that outlines the eastern margin of the Phuquoc–Kampot Som Basin.

faults dominate the deformational belts and occur in imbricate fault systems documented offshore by seismic data. This fault style suggests regional east–west to ENE–WSW compression.

Subordinate NW–SE- to WNW–ESE-trending contractional faults have been mapped in the offshore region (Fig. 5). Many of these faults are remarkably steep and in places form prominent palm structures in combination with faults trending more to north–south and NNW–SSE, suggestive of sinistral transpression along the *c*. NW–SE-trending faults (Fig. 8). Similar trending faults (an order of magnitude smaller) have been studied onshore and on islands in the Gulf of Thailand. A sinistral transpressional component is suggested by these faults in harmony with east–west to ENE–WSW compression.

Timing of inversion

The flat-lying Quaternary to Recent deposits capping the Phuquoc group onshore provide little constraint on the timing of basin inversion. However, interpretation of offshore seismic data provides a better age control. Distinct compressional faults and folds terminate at the angular top-Mesozoic unconformity. These structures strongly influence the depth of truncation along the unconformity but leave the stratigraphic Upper Jurassic to Lower Cretaceous thicknesses unaffected. This demonstrates that the compressional tectonic event post-dates the Early Cretaceous (Aptian) and predates the late Eocene–Recent overburden. The Aptian age of the upper Phuquoc group in Vietnam may be a conservative estimate of the youngest age of the basin fill, as a substantial section has been removed by erosion, further decreasing the potential period of inversion.

Apatite fission track analyses were carried out on seven samples from the Kampot Fold Belt in Vietnam, six from outcrops and one from a well core, to date the inversion more accurately using the approach described by Japsen *et al*. (2007) (Fig. 3, Table 1). Sampling was carried out to cover various stratigraphic levels across a wide area, to optimize the age estimate of the inversion. Igneous rocks studied by AFTA were dated radiometrically (U/Pb on zircon; analytical method described by Frei & Gerdes (2009)) to discriminate between cooling associated with magma solidification and subsequent cooling events more probably caused by exhumation.

The quality of the AFTA data is generally very high, reflecting the excellent apatite yield in most samples. The resulting thermal history is well defined, and overall is regarded as reliable, displaying a high level of consistency between the seven samples of varying lithologies. Fission track ages in all samples are significantly less than the ages of the host rocks, implying post-formational annealing, which, in the light of the seismic analysis, is most probably a result of deeper burial.

Apatite fission track ages in six of the seven samples are similar, ranging from 51.3 ± 3.6 Ma to 62.7 ± 5.9 Ma (Table 2) whereas a single sample of a Triassic sandstone gave an older age of 89.6 ± 12.6 Ma. Mean track lengths are generally between 13 and 14 μm, and track length distributions are broad, with standard deviations generally around 2 μm and a significant proportion of tracks with lengths down to *c*. 10 μm.

Thermal history solutions have been extracted from the AFTA data in these samples following the procedures outlined by Japsen *et al.* (2007). Most importantly, we do not try to constrain the entire thermal history of the sample. Rather we focus on deriving estimates of the maximum palaeotemperature in single samples and the time at which the sample began to cool from the palaeo-thermal maximum (as this is the factor that largely governs the AFTA data).

As summarized in Table 2, all samples show consistent evidence of cooling that began in the interval 62–50 Ma (Palaeocene–early Eocene), and most samples also show evidence of a subsequent late Eocene to early Miocene cooling episode. The sample of a Triassic sedimentary unit (sample 453023) that gave the older fission track age also preserves

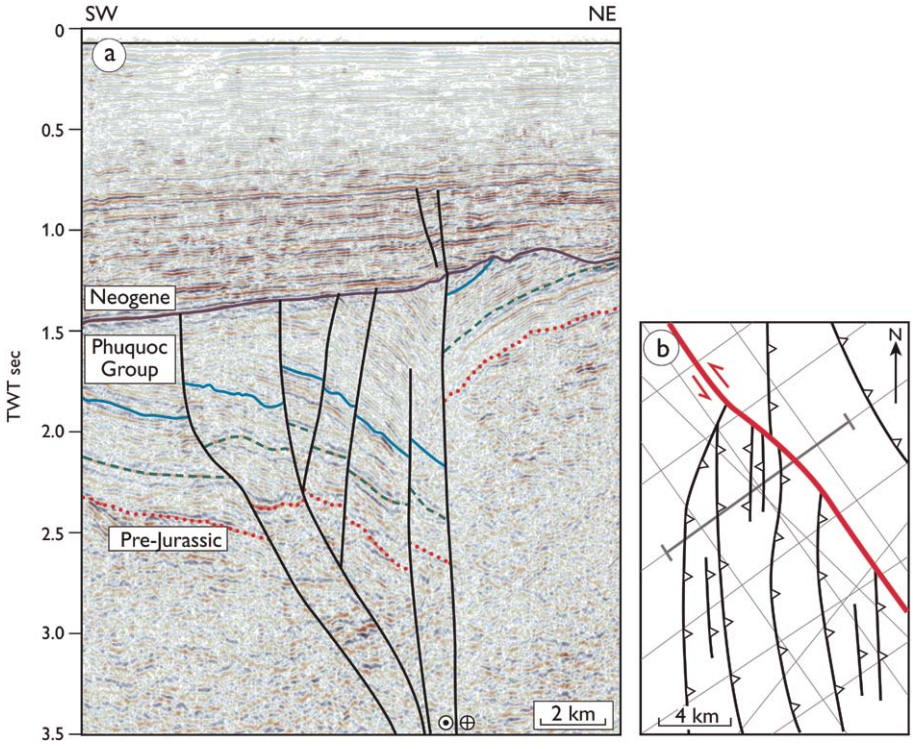


Fig. 8. (**a**) Seismic transect across a positive flower structure, and (**b**) structural map showing fault outline that together with the seismic transect indicates left-lateral transpression across the NW–SE-trending main fault. The seismic grid is shown in (**b**) with the transect emphasized in bold.

Table 2. *Sample details and palaeotemperature analysis combined with stratigraphic and radiometric ages*

Sample number	Latitude (°N)	Longitude (°E)	Rock type	Stratigraphic or radiometric age (Ma)	Present temperature (°C)	AFT age (Ma)*	P (χ^2) %	Mean track length (μm)*	Max. Palaeotemperature, episode 1 (Ma)	Onset of cooling episode 1 (°C)	Maximum palaeotemperature, episode 2 (Ma)	Onset of cooling episode 2 (°C)	Maximum palaeotemperature, episode 3 (Ma)	Onset of cooling episode 3 (°C)
453001†	10°02′48″	103°59′59″	Sandstone	125–112	30	51.3 ± 3.6 (20)	2.6	13.23 ± 0.19 (109)			110–120	70–50	65–80	40–15
453009	09°40′33″	104°23′58″	Meta-sandstone	540–250	20	62.7 ± 5.9 (12)	53.0	12.94 ± 0.49 (14)			>125	90–25	35–120	50–0
453011	09°40′28″	104°21′37″	Tuffite	279 ± 2										
453014	09°41′15″	104°20′59″	Tuffite	277 ± 2										
453015	09°38′12″	104°23′38″	Rhyolite	278 ± 2										
453017	10°01′20″	104°32′50″	Granite	94.9 ± 0.5	20	58.8 ± 3.8 (20)	12.1	13.93 ± 0.18 (111)			>125	60–45	75–85	40–20
453023	10°02′09″	104°32′53″	Sandstone	250–200	20	89.6 ± 12.6 (20)	0.9	13.3 ± 0.55 (29)	>120	130–70	90–100	80–30		
453027	10°02′04″	104°32′49″	Granite	95.4 ± 0.6	20	59.1 ± 7.9 (11)	8.1	13.26 ± 0.32 (43)			>110	80–35	60–85	40–10
453029	10°09′05″	104°54′12″	Granite	98.3 ± 0.6	20	52.8 ± 3.3 (20)	90.9	13.71 ± 0.18 (112)			>105	65–35	55–100	45–20
453030	10°06′16″	104°53′50″	Granodiorite	154‡	20	53.4 ± 2.7 (20)	26.9	13.61 ± 0.17 (107)			>125	55–35	50–95	35–5
								Timing overlap (Ma)§		130–70		55–50		35–20

* Numbers in parentheses following fission track age and mean track length represent the number of single grain ages and track lengths measured.
† Well core sample with a vitrinite reflectance of 0.62 measured slightly above the sample depth.
‡ Vietnamese single-grain K/Ar age after Hoa (1996).
§ Combined timing estimates, assuming that data from all samples represent the effects of synchronous cooling episodes.

evidence of earlier cooling that began between 130 and 70 Ma. This probably records cooling after the emplacement of a Late Cretaceous granite (U/Pb 95.4 ± 0.6 Ma), intruded into the Triassic succession a few tens of metres away from the sample site. The granite was sampled at the same outcrop (sample 453027) and the AFTA data show that this sample cooled below 125 °C during the early Palaeogene episode. The difference in thermal histories between these samples remains unexplained, but the majority of samples provide highly consistent evidence of cooling from palaeotemperatures in excess of 100 °C during early Palaeogene time (between 62 and 50 Ma).

Measured vitrinite reflectance levels (R_o) from 0.59 to 0.63% in the ENRECA II well at slightly shallower depths than the AFTA sample (sample 453001) suggest maximum palaeotemperature in the range 97–104 °C using the Burnham & Sweeney (1989) kinetic model, and following the methods of Japsen *et al.* (2007). This is broadly consistent with the maximum palaeotemperature between 110 and 120 °C derived from AFTA data in the core sample. This confirms that the sampled Early Cretaceous sedimentary unit began to cool from its post-depositional maximum in the early Palaeogene, with cooling beginning between 62 and 50 Ma based on data from all samples (Table 1) taken from units of various stratigraphic ages across a widespread area. The cooling episode probably stems from exhumation during the early Palaeogene. Such an episode is consistent with independent age constraints of the Phuquoc–Kampot Som Basin inversion provided by Cretaceous and middle Cenozoic deposits that bracket the inversion unconformity, as indicated by offshore seismic data and information from wells.

The second episode of cooling, which began between 35 and 20 Ma (latest Eocene to earliest Miocene), probably reflects a second phase of increased denudation (Fig. 9). This phase occurred coevally with regional rifting in the adjacent Khmer and Malay basins as well as along the east coast of Vietnam. Consequently, the second cooling event is interpreted as a result of enhanced

Fig. 9. Schematic illustration of the thermal history of the Phuquoc–Kampot Som Basin exemplified by the ENRECA II well-core sample taken at 496.5–496.75 m in the well. Heating occurred during the Jurassic to Cretaceous burial phase as indicated by apatite fission track annealing and vitrinite data. Subsequent Palaeocene–early Eocene basin inversion is indicated by seismic data and AFTA ages. A second cooling event took place from late Eocene to Oligocene time and was probably caused by uplift along the flanks of the adjacent rift basins.

exhumation of the uplifted basin flanks. Indeed, the truncational pattern towards the Khmer and the Malay basins indicates significant rift-shoulder uplift along the flanks of the main rifts, followed by Neogene subsidence and deposition (Fig. 7).

Regional correlation and linkage of deformational belts

The Khmer–Chanthaburi–Phetchabun Fold Belt

The western boundary of the Phuquoc–Kampot Som Basin can be traced from the offshore Khmer Fold Belt in the central part of the Gulf of Thailand to the Chanthaburi Fold Belt onshore southeasternmost Thailand. The onshore western boundary of the Phuquoc–Kampot Som Basin mirrors the eastern equivalent in terms of topography and overall geological composition. As in the easternmost part of the Phuquoc–Kampot Som Basin, remarkable linear ridges, a few hundred metres high, parallel the deformation belt in the westernmost part of the basin. Seismic data show that a similar ridge pattern exists offshore in the westernmost part of the basin, buried below the Cenozoic sediments.

The Chanthaburi Fold Belt crops out onshore as erosional remnants of deformed Triassic and older rocks, comparable with those sub-cropping in the top-Mesozoic unconformity as observed in the offshore Khmer Fold Belt. The hills are surrounded by Quaternary alluvium, which renders detailed structural mapping difficult (Vimuktanandana 1985; Morley 2002). However, a prominent NNW–SSE-trending fabric is evident in the Chanthaburi Fold Belt, similar to that of the offshore fold belt. This suggests that the Khmer Fold Belt forms the offshore continuation of the Chanthaburi Fold Belt, both together constituting the western erosional boundary of the Phuquoc–Kampot Som Basin, and that the deformational belt stretches more than 500 km from southeastern Thailand to the central Gulf of Thailand. The trace of the belt is lost below Quaternary alluvium near the area where the Mae Ping Fault Zone is predicted to enter Cambodia.

Sone & Metcalfe (2008) regarded the Chantaburi Fold Belt as the southward continuation of the Sukhothai Fold Belt of northwestern Thailand, chiefly based on the distribution of Triassic I-type granites. However, we suggest a correlation with the Phetchabun Fold Belt farther to the east, which delineates the western erosional boundary of the Khorat Basin and can be traced as far south as immediately north of the proposed trace of the Mae Ping Fault Zone, buried underneath Quaternary alluvium. Widespread Triassic granites similar to those of the Sukhothai Fold Belt have been reported from the Phetchabun Fold Belt (Beckinsale *et al*. 1979; Charusiri *et al*. 1993; Stokes *et al*. 1996), and the Late Mesozoic–earliest Cenozoic development and configuration of the Khmer–Chanthaburi Fold Belt and the Phetchabun Fold Belt seems remarkably similar. Both belts strike in a NNW–SSE to NNE–SSW direction, were exhumed during the Palaeocene to middle Eocene, and include deformed Cretaceous sediments associated with east–west compression (Mouret *et al*. 1993; Heggemann *et al*. 1994; Lovatt Smith *et al*. 1996; Stokes *et al*. 1996; Upton 1999; Morley 2004; Morley *et al*. 2007). Moreover, both belts mark the present western erosional margins of a probably once connected Khorat–Phuquoc–Kampong Som Basin.

The Mae Ping Fault Zone

The trace of the Mae Ping Fault Zone is poorly confined in eastern Thailand and even more so farther to the east in Cambodia. The most commonly assumed fault path strikes towards the Tongle Sap Lake and farther to the SE towards the Mekong Delta and the margin of the South China Sea (e.g. Lacassin *et al*. 1997; Morley 2002, 2004; Morley *et al*. 2007; Smith *et al*. 2007). The combination of sporadically outcropping Triassic and older rocks between outcrops of the Phuquoc and Khorat groups resembles the subcrop pattern mapped along the top-Mesozoic unconformity in the offshore fold belt areas, although widespread late Neogene alluvium effectively conceals most of the pre-Tertiary units. This indicates a comparable erosional setting to that of the adjacent fold belts, where the Phuquoc and the Khorat groups have been removed as a result of orogenic uplift along the NW–SE-trending Mae Ping Fault Zone. This supports the suggestion that there was originally a united Khorat–Phuquoc–Kampong Som Basin. The uplift may very well have occurred in response to Palaeocene–early Eocene left-lateral transpression. Regional sinistral transpression along smaller-scale, similar trending faults within the study area supports this inference, as does the evidence for early Palaeogene cooling of Khorat Group sediments sampled from the southwestern part of the basin (Upton 1999). Indeed, the Kampot Fold Belt continues as far north as near to the alleged trace of the Mae Ping Fault Zone, where they may merge. This could indicate a close relation between structural shortening across the Kampot Fold Belt and Palaeocene–early Eocene sinistral transpression across the Mae Ping Fault Zone.

Between *c*. 50 and 100 km of left-lateral displacement seems to have occurred across the Mae Ping Fault Zone during the Cenozoic, as indicated by the offset Phetchabun Fold Belt relative to the Khmer–Chanthaburi Fold Belt. Although the Quaternary cover impedes a more accurate estimate, the offset deformation belt provides one of the most reliable markers with which to evaluate the total offset across the Mae Ping Fault Zone. A left-lateral offset of 50–100 km is compatible with recent estimates of Smith *et al*. (2007), suggesting a 10–30 km offset during the Oligocene, and an unconstrained offset prior to this. Lacassin *et al*. (1993) interpreted at least 35–45 km of left-lateral movement throughout the life span of the Mae Ping Fault Zone based on extrapolation from boudin trails. However, those workers inferred an offset of *c*. 160 km following Tapponnier *et al*. (1986), based on the offset western granite belt of Thailand.

The Three Pagodas Fault Zone

The NW–SE-trending Three Pagodas Fault Zone transects Thailand and splays into loosely defined strands as it approaches the Gulf of Thailand (Morley 2002). One of the splays has been interpreted to bend to the SSE and enter the gulf in the area where the Khmer–Chanthaburi Fold Belt continues onshore or immediately to the west. This suggests a close connection between the Khmer Fold Belt and the left-lateral Three Pagodas Fault Zone, and may indicate a pre-late Eocene transpressional history of the left-lateral Three Pagodas Fault Zone as argued by Morley (2004).

Regional orogenic control

Suturing along west Sundaland

Left-lateral faulting across the Mae Ping Fault Zone has generally been attributed to escape tectonics associated with the Himalayan orogeny and more recently to the accretion of western Myanmar onto SE Asia, and along the Klong Marui and the Rangong faults in Thailand to Cretaceous subduction processes along the western margin of the SE Asia (Watkinson *et al*. 2008). Consequently, left-lateral faulting has been regarded as a

largely middle Eocene to Oligocene or Late Cretaceous–Oligocene phenomenon (Tapponnier *et al*. 1986; Leloup *et al*. 1995, 2001; Lacassin *et al*. 1997; Morley 2004). Likewise, folding and faulting within Mesozoic rocks has been attributed to the same plate-scale events along the western margin of Sundaland. Mouret *et al*. (1993) assigned a Palaeocene age to the onset of a later phase of folding and related it to exhumation along the Phu Phan Uplift in the Khorat Basin. Based on westward intensified deformation, the forcing mechanism was sought to the west and was associated with the accretion of western Myanmar, or alternatively with the subsequent northward indentation of India. A comparable interpretation was favoured by Upton (1999) to explain the regional Palaeocene–Eocene exhumation of the Khorat Basin and the Phetchabun Fold Belt.

Suturing along east Sundaland; the Luconian orogeny

The Palaeocene to early Eocene Phuquoc–Kampot Som Basin inversion most probably formed part of the same regional inversion event noted by Mouret *et al*. (1993) and Upton (1999). However, correlation of the Khorat Basin inversion with plate-scale events to the west may be premature, as comparable deformation along the Kampot Fold Belt took place to the east of the then united Khorat–Phuquoc–Kampong Som Basin. Hence, the deformation was concentrated along well-defined fold belts outlining more stable crustal blocks, and may not contain conclusive evidence as to the fundamental mechanism.

The coeval timing of the exhumation of the Khorat Basin and the Phetchabun Fold Belt farther to the north (Mouret *et al*. 1993; Upton 1999), in addition to the uplift of central Vietnam located along the line of the Phu Phan Uplift noted by Carter *et al*. (2000), suggests that the event influenced a very large region. Farther to the south, on the Malaysian Peninsula, evidence of a comparable basin inversion exists, although its timing and extent remain poorly constrained (Harbury *et al*. 1990).

The regional nature of the inversion event could indicate that it was the result of suturing along the Sundaland margin as suggested by Mouret *et al*. (1993). In contrast to the model of Mouret *et al*. (1993), the Palaeocene cessation of arc-related magmatism in Vietnam around the onset of inversion may provide a clue to the fundamental mechanism of inversion (Table 1). The position and trend of the Kampot Fold Belt relative to the adjacent magmatic arc supports such a link.

The early Palaeogene cessation of arc-related magmatism in the region marks the breakdown of the Pacific subduction underneath Indochina and western Borneo. At around the same time, the Luconian Block accreted onto Sundaland as recorded by the Sarawak orogeny, which is particularly evident in NW Borneo (Benard *et al*. 1990; Hutchison 1996), suggesting a direct link to continental suturing along the eastern margin of Sundaland (Fig. 10). In Borneo, deformation of early Eocene and older sediments and metamorphic rocks, overlain by less deformed middle or late Eocene to Oligocene deposits and volcanic rocks suggests an early to middle Eocene age for the orogeny in Sarawak. This is compatible with, although slightly diachronous to the age of the orogeny in Indochina and Thailand. However, Hutchison (1996) suggested a Palaeocene onset of collision in Borneo, in harmony with the observations from Vietnam.

The Luconian orogeny has received little attention with respect to the coeval tectonic development of Sundaland. Instead, Palaeocene and early Eocene deformation and exhumation of central Sundaland have been linked to accretion events and subduction along the western margin of SE Asia. The apparent linkage of the Kampot and the Khmer fold belts of Luconian affinity with those of central Thailand suggests a close connection between the suturing of Luconia and the establishment (or

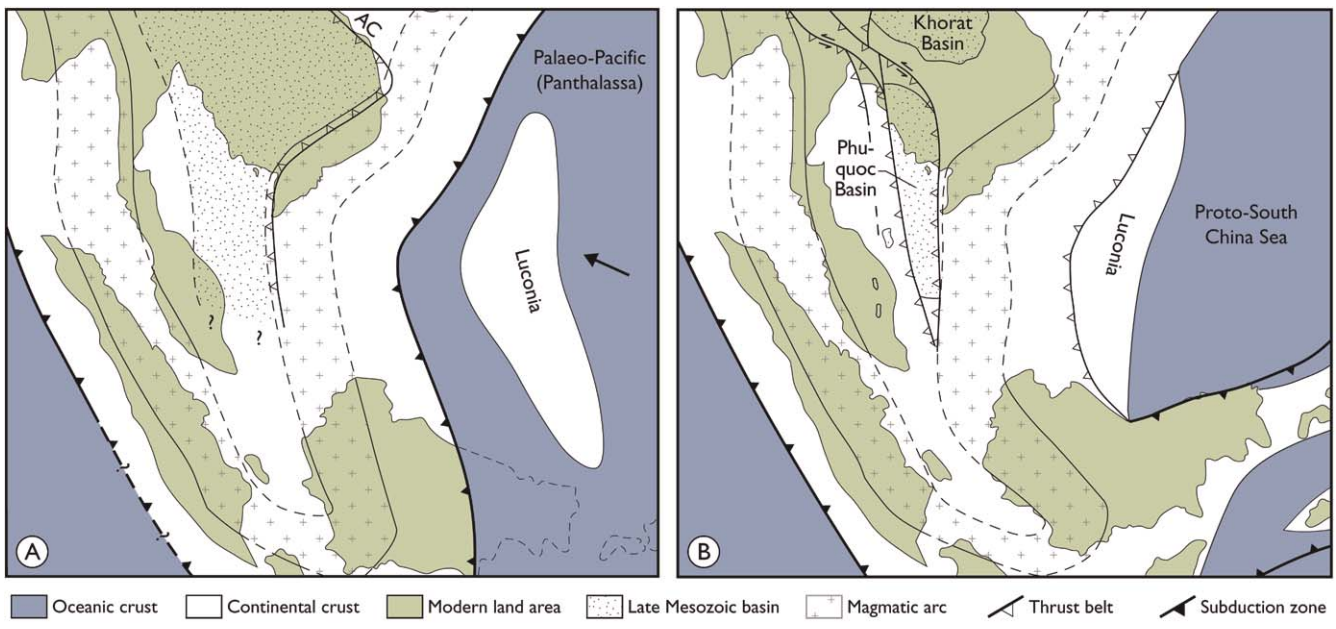


Fig. 10. Simplified reconstruction of the palaeogeographical outline of SE Asia (**a**) immediately before the accretion of Luconia to SE Asia and (**b**) immediately after accretion. Prior to the Luconian suturing a large epicontinental foreland basin formed in association with the rise of a magmatic arc behind the subducting palaeo-Pacific Ocean and uplift along the Annam Cordillera (AC) farther north. Basin inversion associated with the Luconian collision resulted in basin segregation as a result of uplift along well-constrained deformation belts. Compressional folding and faulting dominated along the roughly north–south-trending fold belts, whereas left-lateral transpression took place along the more NW–SE-trending deformation belts.

reactivation) of fold belts, basin inversion, and exhumation in central Sundaland. Moreover, the direct link between the Kampot and the Khmer fold belts and the Mae Ping and the Three Pagodas fault zones, combined with their apparent overlapping timing, suggests that incipient transpression could have been forced by Luconian suturing along the opposite, eastern Sundaland margin. Indeed, left-lateral transpression along parallel NW–SE-trending faults in the Kampot and the Khmer fold belts supports this inference.

Reactivation of crustal structures

Structuring around the Phuquoc–Kampot Som Basin and the Khorat Basin is concentrated along distinct deformational belts. The Phetchabun and the Chanthaburi fold belts have been viewed as Indosinian (Permo-Triassic) structural belts parallel to the Nan Suture (Helmcke 1986; Sone & Metcalfe 2008). The early Cenozoic exhumation and intense structuring of the combined Phetchabun–Khmer Fold Belt are consequently viewed as a reactivation of the Indosinian fold belt. This demonstrates the importance of inherited weakened crustal belts flanking more rigid blocks in the distribution of intra-plate deformation. An equivalent reactivation history may be suspected for the Kampot Fold Belt, which strikes almost parallel to the Indosinian suture zones.

The late Eocene–Oligocene rift system in the Gulf of Thailand has been suggested to follow zones of weakness (Kornsawan & Morley 2002; Morley *et al*. 2004). Extensional reactivation of earliest Cenozoic contractional faults and the concentration of rifting along the Khmer Fold Belt document that middle Cenozoic rifts in the eastern part of the Gulf of Thailand reactivated Palaeocene–early Eocene structural zones that in turn follow the trace of Permo-Triassic deformational belts.

Conclusion

The Phuquoc–Kampot Som Basin in southwestern Indochina forms a Late Jurassic to Early Cretaceous retroarc foreland basin associated with plate convergence and back-arc magmatism along the eastern Sundaland margin. At the time of formation, the Phuquoc–Kampot Som Basin was part of a larger Sundaland basin that included the Khorat Basin located to the north and continued south into Malaysian territories. Non-marine deposition prevailed in the basin, although occasional marine incursions occurred and up to >4 km of sediments accumulated during the period.

Basin inversion occurred during the Palaeocene–early Eocene in response to the Luconian suturing onto SE Asia, which also resulted in basin splitting. The continental accretion affected a large part of Sundaland from Laos in the north to Peninsular Malaya in the south. Along the margin of the Phuquoc–Kampot Som Basin, thrusting and uplift were concentrated within the several hundred kilometre long Kampot and Khmer–Chanthaburi fold belts and sinistral transpression took place across local NW–SE-trending faults. The two fold belts appear to link up with the Mae Ping and Three Pagodas fault zones, suggesting a connection between the onset of left-lateral transpression across Sundaland, Palaeocene–early Eocene basin inversion, and the accretion of Luconia onto SE Asia. The separation of the once-continuous Khmer–Chanthaburi–Phetchabun Fold Belt in Thailand provides a reliable offset geological marker, suggesting a Cenozoic left-lateral offset of 50–100 km across the Mae Ping Fault Zone.

This study was funded by the University of Copenhagen. Additional funding was obtained through the Danida-sponsored ENRECA project and a Geocenter Copenhagen grant. We thank PetroVietnam and Vietnam Petroleum Institute for providing seismic reflection and well data, and for permission to publish this paper. The Geological Survey of Denmark and Greenland (GEUS) is acknowledged for providing facilities for data interpretation. J. Halskov and L. C. Mai are acknowledged for technical assistance. The kind suggestions of E. Sheldon and C. Pulvertaft helped improve an earlier version of the manuscript, and the reviews of C. K. Morley and M. Tingay helped strengthen argumentation.

References

Areshev, E.G., Dong, T.L., San, N.T. & Shnip, O.A. 1992. Reservoirs in fractured basement on the continental shelf of southern Vietnam. *Journal of Petroleum Geology*, **15**, 451–464.

Barber, A.J. & Crow, M.J. 2009. Structure of Sumatra and its implications for the tectonic assembly of Southeast Asia and the destruction of Paleotethys. *Island Arc*, **18**, 3–20.

Barley, M.E., Pickard, A.L., Zaw, K., Rak, P. & Doyle, M.G. 2003. Jurassic to Miocene magmatism and metamorphism in the Mogok metamorphic belt and the India–Eurasia collision in Myanmar. *Tectonics*, **22**, doi:10.1029/2002TC001398.

Beckinsale, R.D., Suensilpong, S., Nakapadungrat, S. & Walsh, J.N. 1979. Geochronology and geochemistry of granite magmatism in Thailand in relation to plate tectonic model. *Journal of the Geological Society, London*, **136**, 529–540.

Benard, F., Muller, C., Letouzey, J., Rangin, C. & Tahir, S. 1990. Evidence of multiphase deformation in the Rajang–Crocker Range (northern Borneo) from Landsat imagery interpretation: geodynamic implications. *Tectonophysics*, **183**, 321–339.

Burnham, A.K. & Sweeney, J.J. 1989. A chemical kinetic model of vitrinite reflectance maturation. *Geochimica et Cosmochimica Acta*, **53**, 2649–2657.

Carter, A. & Bristow, C.S. 2003. Linking hinterland evolution and continental basin sedimentation by using detrital zircon thermochronology: a study of the Khorat Plateau Basin, eastern Thailand. *Basin Research*, **15**, 271–285.

Carter, A. & Moss, S.J. 1999. Combined detrital-zircon fission-track and U–Pb dating: A new approach to understanding hinterland evolution. *Geology*, **27**, 235–238.

Carter, A., Roques, D. & Bristow, C. S. 2000. Denudation history of onshore Central Vietnam: constraints on the Cenozoic evolution of the western margin of the South China Sea. *Tectonophysics*, **322**, 265–277.

Charusiri, P., Clark, A.H., Farrar, E., Archibald, D. & Charusiri, B. 1993. Granite belts in Thailand: evidence from the $^{40}Ar/^{39}Ar$ geochronological and geological syntheses. *Journal of Southeast Asian Earth Sciences*, **8**, 127–136.

Cooper, M.A., Herbert, T. & Hill, G.S. 1989. Stratigraphy of the Huai Hin Lat Formation (Upper Triassic) intermontane basins in northeastern Thailand. *In:* Thanasuthipitak, T. & Ounchanum, P. (eds) *International Symposium on Intermontane Basins: Geology and Resources, Chang Mai, Thailand*. University of Chang Mai, Thailand, 231–242.

DeCelles, P.G. & Giles, K.A.1996. Foreland basin systems. *Basin Research*, **8**, 105–123.

Dien, P.T., Socheat, C., Nielsen, L.H. & Nghinh, L.T. 2008. Indosinian events and petroleum potential of the Phuquoc–Kampong Som area. *In:* Trung, N.H. & Truong, P.V. (eds) *Vien Dau Khi Viet Nam 30, Nam Phat Trien va Hoi Nhap*. Hanoi, Vietnam, 305–319 [in Vietnamese].

Engebretson, D.C., Cox, A. & Gordon, R.G. 1985. *Relative motions between oceanic and continental plates in the Pacific*. Geological Society of America, Special Papers, **206**.

Frei, D. & Gerdes, A. 2009. Precise and accurate *in situ* U–Pb dating of zircons with high sample throughput by automated LA-SF-ICP-MS. *Chemical Geology*, **261**, 261–270, doi:10.1016/j.chemgeo.2008.07.025.

Fuller, M., Ali, J.R., Moss, S.J., Frost, G.M., Richter, B. & Mahfi, A. 1999. Paleomagnetism of Borneo. *Journal of Asian Earth Sciences*, **17**, 3–24.

Fyhn, M.B.W., Boldreel, L.O. & Nielsen, L.H. 2009*a*. Development of the central and south Vietnamese margin: Implications for the establishment of the South China Sea, Indochinese escape tectonics and Cenozoic volcanism. *Tectonophysics*, **478**, 184–224, doi:10.1016/j.tecto.2009.08.002.

Fyhn, M.B.W., Boldreel, L.O. & Nielsen, L.H. 2009*b*. Escape tectonism in the Gulf of Thailand: Paleogene left-lateral pull-apart rifting in the Vietnamese part of the Malay Basin. *Tectonophysics*, doi: 10.1016/j.tecto.2009.11.004, in press.

Fyhn, M.B.W., Nielsen, L.H., Boldreel, L.O., *et al*. 2009*c*. Geological evolution, regional perspectives and hydrocarbon potential of the northwest Phu Khanh Basin, offshore Central Vietnam. *Marine and Petroleum Geology*, **26**, 1–24, doi:10.1016/j.marpetgeo.2007.07.014.

Gobett, D.J. & Hutchison, C.S. (eds) 1973. *Geology of the Malay Peninsula (west Malaysia and Singapore)*. Wiley–Interscience, New York.

Gustavson Associates 1991. *Geology and petroleum resource potential of Laos and Cambodia*. Unpublished report. Gustavson Associates, Boulder, Colorado.

Haile, N.S. 1968. Geosynclinal theory and organization pattern of the NW Borneo Geosyncline. *Quarterly Journal of the Geological Society of London*, **124**, 171–195.

Haile, N.S. 1974. Borneo. *In:* Spencer, A.M. (ed.) *Mesozoic–Cainozoic Orogenic Belts: Data for Orogenic Studies*. Geological Society, London, Special Publications, **4**, 333–347.

Haile, N.S. 1994. The Rajang accretionary prism and the Tran-Borneo Danau suture. *In: Tectonic Evolution of SE Asia. Conference Abstracts Volume, London, 7–8 December 1994*, 17.

Haile, N.S., McElhinny, M.W. & McDougall, I. 1977. Palaeomagnetic data and radiometric ages from the Cretaceous of west Kalimantan (Borneo), and their significance in interpreting regional structure. *Journal of the Geological Society, London*, **133**, 133–144.

Hall, R. 2009. Hydrocarbon basins in SE Asia: understanding why they are there. *Petroleum Geoscience*, **15**, 131–146.

Hall, R., Clements. B. & Smyth, H.R. 2009. Sundaland: Basement character, structure and plate tectonic development. 33rd Annual Convention & Exhibition, May 2009. *Proceedings, Indonesian Petroleum Association*, **IPA09-G-134**.

Hamilton, W. 1979. *Tectonics of the Indonesian Region*. US Geological Survey, Professional Papers, **1078**.

Harbury, N.A., Jones, M.E., Audley-Charles, M.G., Metcalfe, I. & Mohamed, K.R. 1990. Structural evolution of Mesozoic peninsular Malaysia. *Journal of the Geological Society, London*, **147**, 11–26.

Hazebroek, H.P. & Tan, D.N.K. 1993. Tertiary tectonic evolution of the NW Sabah continental margin. *Geological Society of Malaysia Bulletin*, **33**, 195–210.

Heggemann, H., Helmcke, D. & Tietze, K.W. 1994. Sedimentary evolution of the Mesozoic Khorat Basin in Thailand. *Zentralblatt für Geologie und Paläontologie, Teil 1*, **11–12**, 1267–1285.

Helmcke, D. 1986. On the geology of Phetchabun Fold belt (Central Thailand)—implications for the evolution of mainland SE Asia. *Geological Society of Malaysia Bulletin*, **19**, 79–85.

Hoa, N.H. (ed.) 1996. *Geological and mineral resources map of Viet Nam, 1:200 000. Map series C-48*. Department of Geology and Minerals of Vietnam, Hanoi [with explanatory note in Vietnamese and English].

Holloway, N.H. 1981. The stratigraphy and relationship of Reed Bank, north Palawan, and Mindoro to the Asian mainland and its significance in the evolution of the South China Sea. *AAPG Bulletin*, **66**, 1355–1383.

Honza, E. & Fujioka, K. 2004. Formation of arcs and backarc basins inferred from the tectonic evolution of Southeast Asia since the Late Cretaceous. *Tectonophysics*, **384**, 23–53.

Honza, E., John, J. & Banda, R.M. 2000. An imbrication model for the Rajang Accretionary Complex in Sarawak, Borneo. *Journal of Asian Earth Sciences*, **18**, 751–759.

Hutchison, C.S. 1973. Tectonic evolution of Sundaland: a Phanerozoic synthesis. *Geological Society of Malaysia Bulletin*, **6**, 61–86.

Hutchison, C.S. 1989. *Geological evolution of South-East Asia*. Oxford Monographs on Geology and Geophysics, **13**.

Hutchison, C.S. (ed.) 1991. *Studies in East Asian tectonics and resources (SEATAR) crustal transects VII: Java–Kalimantan–Sarawak–South China Sea*. CCOP Technical Publication, **26**.

Hutchison, C. S. 1992. Discussion on structural evolution of Mesozoic Peninsular Malaysia. *Journal of the Geological Society, London*, **149**, 679–680.

Hutchison, C.S. 1996. The 'Rajang accretionary prism' and 'Lupar Line' problem of Borneo. *In:* Hall, R. & Blundell, D. (eds) *Tectonic Evolution of Southeast Asia*. Geological Society, London, Special Publications, **106**, 247–261.

Hutchison, C.S. 2005. *Geology of North-West Borneo, Sarawak, Brunei and Sabah*. Elsevier, Amsterdam.

Jahn, B.M., Zhou, X.H. & Li, J.L. 1976. Rb–Sr ages of granitic rocks in southeastern China and their tectonic significance. *Geological Society of America Bulletin*, **86**, 763–776.

Japsen, P., Green, P.F., Nielsen, L.H., Rasmussen, E.S. & Bidstrup, T. 2007. Mesozoic–Cenozoic exhumation events in the eastern North Sea Basin: a multi-disciplinary study based on palaeothermal, palaeoburial, stratigraphic and seismic data. *Basin Research*, **19**, 451–490.

Katili, J.A. 1973. Geochronology of west Indonesia and its implication on plate tectonics. *Tectonophysics*, **19**, 195–212.

Khoo, H.P. 1977. The geology of the Sungai Tekai area. *Annual Report of Geological Survey of Malaysia*, 93–103.

Koopmans, B.N. 1968. The Tembeling Formation—a lithostratigraphic description (West Malaysia). *Bulletin of the Geological Society of Malaysia*, **1**, 23–43.

Koppers, A.A.P., Morgan, J.P., Morgan, J.W. & Staudigel, H. 2001. Testing the fixed hotspot hypothesis using $^{40}Ar/^{39}Ar$ age progression along seamounts trails. *Earth and Planetary Science Letters*, **185**, 237–252.

Kornsawan, A. & Morley, C.K. 2002. The origin and evolution of complex transfer zones (graben shifts) in conjugate fault systems around the Funan Field, Pattani basin, Gulf of Thailand. *Journal of Structural Geology*, **24**, 435–449.

Lacassin, R., Leloup, P.H. & Tapponnier, P. 1993. Bounds on strain in large Tertiary shear zones of SE Asia from boudinage restoration. *Journal of Structural Geology*, **15**, 677–692.

Lacassin, R., Maluski, H., Leloup, P.H., *et al.* 1997. Tertiary diachronic extrusion and deformation of western Indochina: Structural and $^{40}Ar/^{39}Ar$ evidence. *Journal of Geophysical Research*, **102**, 10013–10037.

Lacassin, R., Leloup, P.H., Trinh, P.T. & Tapponnier, P. 1998. Unconformity of red sandstones in north Vietnam: field evidence for Indosinian Orogeny in northern Indochina. *Terra Nova*, **10**, 106–111.

Lasserre, M., Cheymol, J., Petot, J. & Saurin, E. 1970. Géologie, chimie et géochronologie du granite de Tasal (Cambodge occidental). *Bulletin du BRGM, Section 2*, **4**, 5–13.

Leloup, P.H., Lacassin, R., Tapponnier, P., *et al.* 1995. The ASRR shear zone (Yunnan, China), Tertiary transform boundary of Indochina. *Tectonophysics*, **251**, 3–84.

Leloup, P.H., Arnaud, N., Lacassin, R., *et al.* 2001. New constraints on the structure, thermochronology and timing of the ASRR shear zone, SE Asia. *Journal of Geophysical Research*, **106**, 6683–6732.

Lepvrier, C., Maluski, H., Vuong, N.V., Roques, D., Axente, V. & Rangin, C. 1997. Indosinian NW-trending shear zones within the Truong Son belt (Vietnam) $^{40}Ar/^{39}Ar$ Triassic ages and Cretaceous to Cenozoic overprints. *Tectonophysics*, **283**, 105–127.

Lepvrier, C., Maluski, H., Tich, V.V., Leyreloup, A., Thi, P.T. & Vuong, N.V. 2004. The Early Triassic Indosinian orogeny in Vietnam (Truong Son Belt and Kontum Massif); implications for the geodynamic evolution of Indochina. *Tectonophysics*, **393**, 87–118.

Lepvrier, C., Vuong, N.V., Maluski, H., Thi, P.T. & Vu, T,V. 2007. Indosinian tectonics in Vietnam. *Comptes Rendus Géoscience*, **340**, 94–111.

Li, X.H., Chung, S.-L., Zhou, H., Lo, C.-H., Liu, Y. & Chen, C.-H. 2004. Jurassic intraplate magmatism in southern Hunan–eastern Guangxi: $^{40}Ar/^{39}Ar$ dating, geochemistry, Sr–Nd isotopes and implications for the tectonic evolution of SE China. *In:* Malpas, J., Fletcher, C.J.N., Ali, J.R. & Aitchison, J.C. (eds) *Aspects of the Tectonic Evolution of China*. Geological Society, London, Special Publications, **226**, 193–215.

Linh, H.T. (ed.) 2003. *Geological map of the Phu Quoc Island, 1:50 000*. Department of Geology and Minerals of Vietnam, Hanoi [in Vietnamese].

Lovatt Smith, P.F. & Stokes, R.B. 1997. Geology and petroleum potential of the Khorat Plateu Basin in the Vientiane area of Lao P.D.R. *Journal of Petroleum Geology*, **20**, 27–50.

Lovatt Smith, P.F., Stokes, R.B., Bristow, C. & Carter, A. 1996. Mid-Cretaceous inversion in Northern Khorat Plateau of Lao PDR and Thailand. *In:* Hall, R. & Blundell, D. (eds) *Tectonic Evolution of Southeast Asia*. Geological Society, London, Special Publications, **106**, 233–247.

Metcalfe, I. 1996. Pre-Cretaceous evolution of SE Asian terranes. *In:* Hall, R. & Blundell, D. (eds) *Tectonic Evolution of Southeast Asia*. Geological Society, London, Special Publications, **106**, 97–122.

Metcalfe, I. 1998. Palaeozoic and Mesozoic geological evolution of the SE Asian region, multidisciplinary constraints and implications for biogeography. *In:* Hall, R. & Holloway, J.D. (eds) *Biogeography and Geological evolution of SE Asia*. Backhuys publishers, Amsterdam, The Netherlands, 25–41.

Mitchell, A.H.G. 1993. Cretaceous–Cenozoic tectonic events in the western Myanmar (Burma)–Assam region. *Journal of the Geological Society, London*, **150**, 1089–1102.

Mitchell, A.H.G., Htay, M.T., Htun, K.M., Win, M.N., Oo,T. & Hlaing, T. 2007. Rock relationships in the Mogok metamorphic belt, Tatkon to Mandalay, central Myanmar. *Journal of Asian Earth Sciences*, **29**, 891–910.

Morley, C.K. 2002. A tectonic model for the Tertiary evolution of strike-slip faults and rift Basins in SE Asia. *Tectonophysics*, **347**, 189–215.

Morley, C.K. 2004. Nested strike-slip duplexes, and other evidence for Late Cretaceous–Palaeogene transpressional tectonics before and during India–Eurasia collision, in Thailand, Myanmar and Malaysia. *Journal of the Geological Society, London*, **161**, 799–812.

Morley, C.K., Haranya, C., Phoosongsee, W., Pongwapee, S., Kornsawan, A. & Wonganan, N. 2004. Activation of rift oblique and rift parallel pre-existing fabrics during extension and their effect on deformation style: examples from the rifts of Thailand. *Journal of Structural Geology*, **26**, 1803–1829.

Morley, C.K., Smith, M., Carter, A., Charusiri, P. & Chantraprasert, S. 2007. Evolution of deformation styles at a major restraining bend, constraints from cooling histories, Mae Ping fault zone, western Thailand. *In:* Cunningham, W.D. & Mann, P. (eds) *Tectonics of Strike-slip Restraining and*

Releasing Bends. Geological Society, London, Special Publications, **290**, 325–349.

Moss, S.J. 1998. Embaluh Group turbidites in Kalimantan: evolution of a remnant oceanic basin in Borneo during the Late Cretaceous to Palaeogene. *Journal of the Geological Society, London*, **155**, 509–524.

Mouret, C., Heggemann, H., Goudain, J. & Krisadasima, S. 1993. Geological history of the siliciclastic Mesozoic strata of the Khorat Group in the Phu Phan Range area, northeastern Thailand. *In:* Thanasuthipitak, T. (ed.) *International Symposium, Biostratigraphy of Mainland Southeast Asia: Facies and Paleontology, 31 January–5 February 1993, Vol. 1*. Chiang Mai, Thailand, 23–49.

My, B.P., Linh, T.H., Giap, K.V. & Kham, H.D. 2002. Red beds in the Tho Chu Archipelago, Kien Giang Province. *Journal of Geology, Series B*, **19–20**. World Wide Web Address: http://www.idm.gov.vn/Nguon_luc/Xuat_ban/Tap-chi2003/SeriesB/19_20/t35.htm.

Naylor, M. & Sinclair, H.D. 2008. Pro- vs. retro-foreland basins. *Basin Research*, **20**, 285–303, doi:10.1111/j.1365-2117.2008.00366.x.

Ngah, K., 2000. Structural framework of southeastern Malay Basin. *Search and Discovery*, Article 10009. World Wide Web Address: http://www.searchand-discovery.net/documents/khalid02/index.htm.

Nguyen, T.T.B., Satir, M., Siebel, W. & Chen, F. 2004. Granitoids in the Dalat zone, southern Vietnam: age constraints on magmatism and regional geological implications. *Geologische Rundschau*, **93**, 329–340.

Omang, S.A.K. & Barber, A.J. 1996. Origin and tectonic significance of the metamorphic rocks associated with the Darvel Bay Ophiolite, Sabah, Malaysia. *In:* Hall, R. & Blundell, D.J. (eds) *Tectonic Evolution of SE Asia*. Geological Society, London, Special Publications, **106**, 263–279.

Pradidtan, S. & Dook, R. 1992. Petroleum geology of northern part of the Gulf of Thailand. *In*: Piancharoen, C. (ed.) *Proceedings of Conference on Geologic Resources of Thailand: Potential for Future Development, Bangkok, Thailand, November 1992*, 235–246.

Racey, A., Love, M.A., Canham, A.C., Goodall, J.G.S., Polachan, S. & Jones, P.D. 1996. Stratigraphy and reservoir potential of the Mesozoic Khorat Group, NE Thailand. *Journal of Petroleum Geology*, **19**, 5–40.

Rangin, C., Huchon, P., Le Pichon, X., *et al.* 1995. Cenozoic deformation of Central and South Vietnam. *Tectonophysics*, **235**, 179–196.

Rishworth, D.E.H. 1974. *The Upper Mesozoic Terrigenous Gagau Group of Peninsular Malaysia*. Geological Survey of Malaysia Special Paper, **1**.

Salvador, A. (ed.) 1994. *International stratigraphic guide: a guide to stratigraphic classification, terminology and procedure*. International Union of Geological Science, IUGS Secretariat, Trondheim; Geological Society of America, Boulder, CO

Schmidtke, E., Fuller, M. & Haston, R. 1990. Paleomagnetic data from Sarawak, Malaysian Borneo and the Late Mesozoic and Cenozoic tectonics of Sundaland. *Tectonics*, **9**, 123–140.

Searle, M.P. & Morley, C.K. in press. Tectonic and thermal evolution of Thailand in the regional context of Southeast Asia. *In:* Ridd, M.F., Barber, A.J. & Crow, M.J. (eds) *Geology of Thailand*. Geological Society, London, (in press).

Searle, M.P., Noble, S.R., Cottle, J.M., Waters, D.J., Mitchell, A.H.G., Hlaing, T. & Horstwood, M.S.A. 2007. Tectonic evolution of the Mogok metamorphic belt, Burma (Myanmar) constrained by U–Th–Pb dating of metamorphic and magmatic rocks. *Tectonics*, **26**, doi:10.1029/2006TC002083.

Seton, M. & Müller, R.D. 2008. Reconstructing the junction between Panthalassa and Tethys since the Early Cretaceous. *In: Eastern Australasian Basins III*. Petroleum Exploration Society of Australia, Special Publications, 263–266.

Smith, A. 2007. A plate model for Jurassic to recent intraplate volcanism in the Pacific Ocean basin. *In:* Foulger, G.R. & Jurdy, D.M. (eds) *Plates, Plumes, and Planetary Processes*. Geological Society of Amarica, Special Papers, **430**, 471–495.

Smith, M., Chantraprasert, S., Morley, C.K. & Cartwright, I. 2007. Structural geometry and timing of deformation in the Chainat duplex, Thailand. *In:* Cunningham, W.D. & Mann, P. (eds) *Tectonics of Strike-slip Restraining and Releasing Bends*. Geological Society, London, Special Publications, **290**, 305–323.

Sone, M. & Metcalfe, I. 2008. Parallel Tethyan sutures in mainland Southeast Asia: New insight for paleo-Tethys closure and implications for the Indosinian orogeny. *Comptes Rendus Géoscience*, **340**, 166–179.

Stokes, R.B., Lovatt Smith, P.F. & Soumphonphakdy, K. 1996. Timing of the Shan–Thai–Indochina collision: new evidence from the Pak Lay Foldbelt of the Lao PDR. *In:* Hall, R. & Blundell, D. (eds) *Tectonic Evolution of Southeast Asia*. Geological Society, London, Special Publications, **106**, 305–323.

Tan, D.N.K. & Lamy, J.M. 1990. Tectonic evolution of the NW Sabah continental margin since the Late Eocene. *Bulletin of the Geological Society of Malaysia*, **27**, 241–260.

Tapponnier, P., Peltzer, G. & Armijo, R. 1986. On the mechanics of the collision between India and Asia. *In:* Coward, M.P. & Ries, A.C. (eds) *Collision Tectonics*. Geological Society, London, Special Publications, **19**, 115–157.

Thang, N.D. (ed.) 1999. *Geological and mineral resources map of Viet Nam, 1:200 000. Map series C 48-49 and D 48-49*. Department of Geology and Minerals of Vietnam, Hanoi [with explanatory note in Vietnamese and English].

Thuy, N.T.B., Satir, M., Siebel, W., Vennemann, T. & Long, T.V. 2004. Geochemical and isotopic constraints on the petrogenesis of granitoids from the Dalat zone, southern Vietnam, *Journal of Asian Earth Sciences*, **23**, 467–467.

Tien, P.C. (ed.) 1991. *Geological Map of Cambodia, Laos and Vietnam. Central Vietnam, Map sheet 1:1 000 000*, 2nd edn. Geological Survey of Vietnam, Hanoi [with explanatory note in Vietnamese and English].

Tinh, T. (ed.) 1998. *Geological and mineral resources map of Viet Nam, 1:200 000. Map series D48–D49*. With explanatory note (in Vietnamese and English). Department of Geology and Minerals of Vietnam, Hanoi.

Tongkul, F. 1991. Structure style and tectonics of western and Northern Sabah. *Geological Society of Malaysia Bulletin*, **27**, 227–239.

Trang, N.V. (ed.) 1998. *Geological and mineral resources map of Viet Nam, 1:200 000. Map series D-49*. Department of Geology and Minerals of Vietnam, Hanoi [with explanatory note in Vietnamese and English].

Tri, T.V. (ed.) 1999. *Geological and mineral resources map of Viet Nam, 1:200 000. Map series D-49*. Department of Geology and Minerals of Vietnam, Hanoi [with explanatory note in Vietnamese and English].

Upton, D.R. 1999. *A regional fission track study of Thailand: Implications for thermal history and denudation*. PhD thesis, University of London.

Vimuktanandana, S. (ed.) 1985. *Geological map of Thailand, Map series 1:250 000*. Geological Survey Division, Department of Mineral Resources, Bangkok.

Vimuktanandana, S. (ed.) 1999. *Geological map of Thailand (1:2 500 000)*. Geological Survey Division, Department of Mineral Resources, Bangkok.

Vysotsky, V.I., Rodinikova, R.D. & Li, M.N. 1994. The petroleum geology of Cambodia. *Journal of Petroleum Geology*, **17**, 195–210.

Watkinson, I., Elders, C. & Hall, R. 2008. The kinematic history of the Khlong Marui and Ranong Faults, southern Thailand. *Journal of Structural Geology*, **30**, 1554–1571.

Williams, P.R., Johnston, C.R., Almond, R.A. & Simamora, W.H. 1988. Late Cretaceous to Early Tertiary structural elements of West Kalimantan. *Tectonophysics*, **148**, 279–297.

Williams, P.R., Supriatana, S., Johnston, C.R., Almond, R.A. & Simamora, W.H. 1989. A Late Cretaceous to Early Tertiary accretionary complex in West Kalimantan. *Bulletin of the Geological Research and Development Centre*, **13**, 9–29.

Zhou, X.M. & Li, W.X. 2000. Origin of late Mesozoic igneous rocks in southeastern China: implications for the lithosphere subduction and underplating of mafic magmas. *Tectonophysics*, **326**, 269–287.

Received 17 March 2009; revised typescript accepted 9 October 2009.
Scientific editing by Alan Collins.

View publication stats